Certified Valuation of Arithmetic Asian Options

Conditional Representation, Geometric Reference and Heston Error Bounds

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Abstract

How much numerical error is introduced when a Heston arithmetic Asian price is computed under a projected Euler law? We answer this question at a fixed time step for a bounded call spread with discrete monitoring, in the form of a signed enclosure for the Euler-minus-continuous price difference. For a one-year, twelve-fixing 95–110 spread at the specified Heston parameters and step $h = 1/768$, this enclosure is

$$[-0.011024692273,\ 0.010642371599],$$

and the continuous price belongs to

$$[6.508371733,\ 6.518868974].$$

The calculation includes variance projection, transform inversion, integration tails and rounding. It rests on three steps. Integrating a Gaussian factor shared by every fixing gives an exact conditional price. The geometric average preserves this factor and has a logarithm that is linear in the observed log prices, making it a tractable reference for the arithmetic payoff. Convexity and an explicit curvature calculation then bound the residual by a second moment weighted by the inverse square root of first-period integrated variance, and validated transforms evaluate the result. The separate-law bound can retain a positive width even for identical laws; this explains why finer time steps alone do not make the certificate arbitrarily sharp. A direct coupling estimate preserves the missing cancellation and gives a narrower remainder bound in a deterministic-variance example. Weak expansions and posterior-quantile transfer are developed separately on their stated parameter families.

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1 Introduction

A valuation system replaces a continuous stochastic-volatility model by a discrete law. Even when sampling error is negligible, the resulting price need not equal the price of the model that was calibrated. For an arithmetic Asian contract, this difference depends on the joint law at all fixing dates and on the kinks of the payoff. We ask for a finite bound, in price units, on the error introduced by one specified implementation: positive-part Euler for the variance and a log-price update using the current variance.

The contract is a bounded arithmetic call spread with a fixed observation schedule. Fixing the schedule separates the time discretization of the dynamics from any approximation of a continuously monitored average. The call spread supplies a deterministic bound on the payoff. The model parameters and the discrete kernel remain fixed throughout the comparison; the question is about implementation error within this model and not about which model describes the market.

For the one-year, twelve-fixing spread studied here, at 768 Euler steps, the answer is a signed interval less than .0217 wide. The projected-Euler price lies within .011025 of the continuous Heston price, and the payoff remainder accounts for about 94.5 percent of the interval's width. The rest of this introduction describes how the interval is built and why it has this width.

The first step is an exact reduction. The independent stock noise accumulated before the first fixing enters every later stock price through a common lognormal multiplier. Conditioning on the variance driver and the remaining stock noise leaves an ordinary lognormal call integral. This integrates out the payoff kink but leaves an expectation over the other conditional variables, and the remaining task is to bound that outer expectation.

The geometric average is a natural reference. It scales in exactly the same way as the arithmetic average under the common Gaussian factor, and its logarithm is a linear combination of the fixing-date log prices. The first property allows a conditional Taylor expansion, and the second allows evaluation by affine transforms. A weighted least-squares criterion characterizes the optimal exact scale and provides a benchmark for assessing the rational scale used in the finite calculation.

Conditional curvature determines the weight in that criterion. If A and G denote the arithmetic and geometric averages and I is the first-period integrated variance, the nonlinear error is controlled by

$$E\Big[(A - cG)^2 I^{-1/2}\Big].$$

The inverse square root appears because less conditional variance gives less smoothing at a strike. Expanding the square produces the finite moment catalog needed for the proof; in the twelve-fixing case it contains 91 real loadings. The linear term requires one distribution transform and twelve stock-weighted transforms. Both catalogs are thus fixed by the contract and the exact decomposition.

Writing $\mathcal{L}$ for the linear reference payoff and $R_{K,\nu}$ for its nonnegative call remainder under law ν, the price difference has the exact form

$$\begin{aligned} e_h = e^{-rT}\big[&(E_{Q_h}\mathcal{L} - E_P\mathcal{L}) \\ &+ (R_{K_1,Q_h} - R_{K_1,P}) - (R_{K_2,Q_h} - R_{K_2,P})\big]. \end{aligned} \tag{1.1}$$

The identity splits the work in two. The transform calculation has to enclose the linear difference under the actual projected kernel, and the remainder argument has to control what is lost when the arithmetic payoff is replaced by its reference. Lemma 3.1 gives the weighted remainder bound, and Theorem 4.1 combines the two parts into a signed interval. Appendices

A–C verify the moment, projection, inversion and tail estimates needed to evaluate that interval.

Separate-law bounds have a built-in limitation. When the two laws coincide, their remainders cancel exactly in (1.1), but separate upper bounds need not cancel. The resulting width can therefore remain positive as the time step decreases. Section 4 identifies this obstruction analytically before examining the numerical grid study. Appendix H derives a direct bound on the difference of the remainders; its deterministic-variance example shows what can be gained by retaining the relation between the laws. The corresponding coupled moments have not been computed at the stochastic-variance Heston points.

Conditioning and geometric reference variables have a well-established role in Asian option pricing [1, 2, 3, 4]. In particular, Fusai and Kyriakou treat Heston Asian options and derive conditional-variance error bounds [4]. Heston's affine structure supplies the continuous transforms [5]; recursive transforms for discrete geometric averages are also available [6, 7]. The geometric reference and the principle of conditioning are classical. The result here concerns the weighted kink remainder and its combination with complete error accounting for the original projected Euler law.

Sections 2 and 3 develop the exact conditional representation and the payoff decomposition. Section 4 states the price certificate and explains its width. Section 5 verifies the common-factor structure for deterministic variance and Heston dynamics. Sections 6 and 7 study weak expansions and posterior transfer on separate parameter families. Section 8 uses the numerical certificates to examine the terms predicted by the analysis; Section 9 records the boundaries of the results.

1.1 Comparison with conditional Asian bounds

Fusai and Kyriakou [4] already obtain optimized conditioning bounds for Heston Asian options (Sections 2.2–3), conditional-variance error bounds (Theorem 4, equations (51)–(54)), and experiments at several monitoring frequencies (Section 5.1, Table 5). The table below shows which mathematical and computational objects are studied here. Our principal result is the finite error enclosure for the original kernel (2.2), built from a payoff remainder weighted by first-period integrated variance and a validated computation of every stated error contribution.

Their conditional-variance argument uses a monotonicity condition on the relevant conditional expectation, which they check numerically in their experiments (Appendix A, p. 22).

Page references in Table 1 use the internal numbering of the accepted manuscript linked in [4], after its repository cover, rather than the journal pagination.

Table 1. Comparison with the conditional Asian bounds of Fusai and Kyriakou.

Object	Fusai–Kyriakou	Present result
Payoff approximation	Optimized conditional lower bound and conditional-variance error (Theorem 4, p. 15).	Exact linear-reference decomposition with nonnegative remainder controlled by a weighted second moment (Lemma 3.1).
Discrete numerical law	QE simulation benchmark (Appendix B, p. 22); no certificate for kernel (2.2) is reported.	Projection correction between an algebraic transform recursion and the specified original positive-part variance Euler law, using current variance in the stock update (Appendix B).
Transform computation	Fourier conditional moments (equations (55)–(56), pp. 15–16) and FFT implementation (Section 5, pp. 16–17).	Finite real and complex loading catalogs, certified quadrature and infinite-tail bounds, and complex-logarithm branch checks (Appendices A–C).
Finite arithmetic	An outward-rounded error budget is not reported in the cited paper.	Rational endpoint combination and ball-arithmetic enclosures include rounding with the other finite errors (Appendices A–C and G).
Certified output	Lower-bound approximation error (Theorem 4); no executable signed Euler-minus-continuous enclosure is reported.	Signed $p_h - p_c$ interval at the specified parameters and grid (Theorem 4.1 and Corollary 4.2), traceable to the accompanying executable package.

Relative to Fusai and Kyriakou, this paper adds the weighted remainder and its combination with error control for the original projected kernel. Section 4 analyzes the width of the certificate that results.

1.2 Three distinct domains of validity

The main financial question concerns a fixed contract. In all monetary Asian conclusions this is the arithmetic call spread in (2.3), whose payoff is bounded by 15. Table 2 distinguishes the domains of the three results. The posterior example concerns a near-deterministic perturbation regime rather than calibration stability at the main Heston point.

Table 2. Domains of the three certified results.

Result	Domain and interpretation
Main price certificate	At (4.3), $h = 1/384, 1/768, 1/1536$; at the separate $v_0 = .04$ point in Section 8.3, only $h = 1/768$. Both use $\xi = .23$, twelve monthly fixings and the original projected Euler kernel.
Ten-payoff weak expansion	Theorem 6.1 under (6.1); numerical family (6.10): $\xi = 0$, $\kappa = 3$, $\bar{v} = .045$, $v_0 \in [.03, .06]$, twelve monthly fixings, aligned $h \le 1/768$.
Posterior quantile transfer	Independent uniform prior (7.5), $\xi \in [10^{-7}, 10^{-6}]$, exact synthetic quotes in Appendix F.

2 The valuation problem and its exact conditional representation

2.1 The continuous model, discrete kernel, and payoff

Let W, B be independent standard Brownian motions, and let $\theta = (\kappa, \bar{v}, \xi, \rho, v_0)$. Assume $S_0, \kappa, \bar{v}, v_0 > 0$, $\xi \ge 0$, $|\rho| < 1$, and $r \in \mathbb{R}$. The discrete model uses $h > 0$, with each t_i/h a positive integer. The continuous model is

$$\begin{aligned} dV_t &= \kappa(\bar{v} - V_t)\,dt + \xi\sqrt{V_t}\,dW_t, \\ dZ_t &= (r - V_t/2)\,dt + \sqrt{V_t}\big(\rho\,dW_t + \sqrt{1-\rho^2}\,dB_t\big), \\ S_t &= S_0 e^{Z_t}, \qquad Z_0 = 0, \quad V_0 = v_0. \end{aligned} \tag{2.1}$$

On the aligned grid with step size h, define the original discrete kernel Q_h by

$$\begin{aligned} V^h_{j+1} &= \Big[(1 - \kappa h)V^h_j + \kappa\bar{v}h + \xi\sqrt{hV^h_j}\,G_j\Big]^+, \\ Z^h_{j+1} &= Z^h_j + (r - V^h_j/2)h + \sqrt{hV^h_j}\big(\rho G_j + \sqrt{1-\rho^2}H_j\big), \end{aligned} \tag{2.2}$$

where G_j, H_j are independent standard normal variables, the two updates at the same step share G_j, and the initial conditions agree. In particular, the price update uses V^h_j.

Fix $0 < t_1 < \cdots < t_n = T$, and set

$$A = \frac{1}{n}\sum_{i=1}^{n} S_{t_i}, \qquad G = \left(\prod_{i=1}^{n} S_{t_i}\right)^{1/n}, \qquad \psi(x) = (x - K_1)^+ - (x - K_2)^+, \tag{2.3}$$

where $0 < K_1 < K_2$. Write

$$p_c = e^{-rT}E_P\psi(A), \qquad p_h = e^{-rT}E_{Q_h}\psi(A), \qquad e_h = p_h - p_c. \tag{2.4}$$

We use this sign convention for the error throughout, and when one step size is understood we write $Q = Q_h$. The numerical example uses $n = 12, t_i = i/12, T = 1$, and

$$S_0 = 100, \quad r = .01, \quad K_1 = 95, \quad K_2 = 110, \quad h = 1/768. \tag{2.5}$$

The calibration payoffs are the nine European puts with maturities $1/4, 1/2, 1$ and strikes $90, 100, 110$.

2.2 The common Gaussian structure H_s

A pricing law ν is said to satisfy $\mathsf{H}_s(c)$ if there exist a sub-σ-algebra $\mathcal{G}$, positive $\mathcal{G}$-measurable random variables $a_1, \dots, a_n, \sigma$, and a random variable U such that

$$U \mid \mathcal{G} \sim N(0, \sigma^2), \qquad S_{t_i} = e^U a_i. \tag{2.6}$$

Set $a = n^{-1} \sum_i a_i$ and $g = (\prod_i a_i)^{1/n}$. We require

$$E_\nu(A + G) < \infty, \qquad D_\nu(c) := E_\nu\left[\frac{(A - cG)^2}{\sigma}\right] < \infty. \tag{2.7}$$

Here $c > 0$ is a constant chosen in advance. Condition (2.6) describes the stochastic structure of the model, and (2.7) is an integrability condition that one can check through moments or Laplace transforms, so neither condition already contains the price-difference conclusion we want to prove.

2.3 Exact conditional valuation

The desired output is a finite interval for the actual price difference,

$$e_h \in [\underline{e}, \overline{e}]. \tag{2.8}$$

To derive it, we first evaluate the common Gaussian factor without approximation. Let Φ and φ denote the standard normal distribution function and density.

Proposition 2.1 (Exact conditional price). Under $\mathsf{H}_s(c)$, define, for $x, s, K > 0$,

$$F_K(x; s) = x e^{s^2/2} \Phi\left(\frac{\log(x/K)}{s} + s\right) - K \Phi\left(\frac{\log(x/K)}{s}\right). \tag{2.9}$$

With $a = n^{-1} \sum_i a_i$ from (2.6), the exact price under either law is

$$p_\nu = e^{-rT} E_\nu \left[F_{K_1}(a; \sigma) - F_{K_2}(a; \sigma)\right]. \tag{2.10}$$

Proof. Conditional on $\mathcal{G}$, write the Gaussian factor as sZ, with Z standard normal. The event $x e^{sZ} > K$ is $Z > \log(K/x)/s$. Completing the square in $e^{sz}\varphi(z)$ gives

$$E[e^{sZ} \mathbf{1}_{\{Z > q\}}] = e^{s^2/2} \Phi(s - q), \qquad P(Z > q) = \Phi(-q).$$

Substitution gives (2.9). Apply it at the two strikes and then take expectations. Integrability follows from (2.7), and the conditional spread itself lies between zero and $K_2 - K_1$. □

Formula (2.10) does the one-dimensional conditional integration exactly, but it leaves the outer expectation over the variance path and the remaining stock noise. This is why the next section introduces an approximation: we look for a reference that transforms can compute and an explicit bound on the residual in this exact representation.

3 A geometric reference and the weighted payoff remainder

3.1 Why the geometric average is a tractable reference

Both averages inherit the common multiplier in (2.6): $A = e^U a$ and $G = e^U g$. The geometric average also satisfies $\log(G/S_0) = n^{-1} \sum_i Z_{t_i}$, so products of its powers with stock prices are exponential-affine loadings of the fixing-date log prices. These two properties connect the conditional price to the available transform formulas.

There is a simple characterization within this class. Suppose a positive monomial reference has the form $C\prod_i s_i^{w_i}$, with $C>0$ and real exponents. Invariance under every permutation of the fixing prices forces all exponents to agree, while degree-one homogeneity forces their sum to be one. Consequently $w_i = 1/n$, and the reference is CG. The statement is about monomials only and says nothing about optimality among all symmetric functions of the prices.

We therefore expand the exact conditional call about cg, with one positive constant used under both laws. Convexity fixes the sign of the residual, and curvature fixes the moment needed to bound it. Section 4.1 characterizes the scale minimizing the exact nonlinear width and compares an enclosure of that optimizer with the rational scale used in the certificate.

3.2 Conditional curvature and the remainder

Lemma 3.1. Suppose that $\mathsf{H}_s(c)$ holds. For $K>0$, define

$$R_{K,\nu} = E_\nu\big[(A-K)^+ - (cG-K)^+ - (A-cG)\mathbf{1}_{\{cG>K\}}\big]. \tag{3.1}$$

Then

$$0 \le R_{K,\nu} \le \frac{D_\nu(c)}{2K\sqrt{2\pi}}. \tag{3.2}$$

Proof. Convexity of the positive-part function gives pathwise nonnegativity. Conditional on $\mathcal{G}$, let

$$F_K(x) = F_K(x;\sigma) = E[(xe^U - K)^+ \mid \mathcal{G}], \qquad x>0.$$

Since $\sigma>0$, this function is twice differentiable, with

$$F_K''(x) = \frac{K}{\sigma x^2}\varphi\!\left(\frac{\log(x/K)}{\sigma}\right) \le \frac{e^{2\sigma^2}}{K\sigma\sqrt{2\pi}}, \tag{3.3}$$

where φ is the standard normal density. The final bound follows by maximizing at $\log(x/K) = -2\sigma^2$.

Applying Taylor's formula with integral remainder between the positive arguments a, cg gives

$$0 \le F_K(a) - F_K(cg) - (a-cg)F_K'(cg) \le \frac{e^{2\sigma^2}(a-cg)^2}{2K\sigma\sqrt{2\pi}}.$$

The conditional derivative satisfies

$$(a-cg)F_K'(cg) = E[(A-cG)\mathbf{1}_{\{cG>K\}} \mid \mathcal{G}].$$

Since

$$E[(A-cG)^2 \mid \mathcal{G}] = e^{2\sigma^2}(a-cg)^2,$$

taking expectations proves (3.2). The call and linear terms are integrable because $E(A+G)<\infty$. The conditional second-moment identity is used in the sense of nonnegative conditional expectations, and the weighted remainder is controlled by $D_\nu(c)<\infty$. □

The proof works because the two factors $e^{2\sigma^2}$ match exactly, which lets us write the curvature contribution from conditional smoothing directly as a weighted second moment under the true model. The residual $A-cG$ may have either sign.

3.3 The payoff decomposition

Define

$$\mathcal{L} = \psi(cG) + (A - cG)\mathbf{1}_{\{K_1 < cG \le K_2\}}. \tag{3.4}$$

The conditional normal variance is strictly positive, so G has no atom at any positive threshold. By (3.1),

$$E_\nu \psi(A) = E_\nu \mathcal{L} + R_{K_1,\nu} - R_{K_2,\nu}. \tag{3.5}$$

The identity keeps the original arithmetic payoff as the object of interest and splits off a linear part that a small number of transforms can compute.

Let $Y = \log(G/S_0)$ and $y_K = \log(K/(cS_0))$, and define the distribution functions of two positive measures by

$$F_{0,\nu}(y) = \nu(Y \le y), \qquad F_{A,\nu}(y) = E_\nu[A\mathbf{1}_{\{Y \le y\}}].$$

Expanding (3.4) on each of its intervals gives

$$E_\nu \mathcal{L} = (K_2 - K_1) + K_1 F_{0,\nu}(y_{K_1}) - K_2 F_{0,\nu}(y_{K_2}) - F_{A,\nu}(y_{K_1}) + F_{A,\nu}(y_{K_2}). \tag{3.6}$$

The corresponding transforms are

$$\widehat{\mu}_0(z) = E_\nu e^{zY}, \qquad \widehat{\mu}_A(z) = \frac{S_0}{n}\sum_{i=1}^{n} E_\nu e^{Z_{t_i} + zY}. \tag{3.7}$$

Thus the linear part requires thirteen loadings when $n = 12$: one for the law of the geometric average and one for each summand of the arithmetic average. The number is fixed by the payoff decomposition.

4 The main theorem and the complete price certificate

Finite verification inputs.

The structural condition must also be connected to finite computation. The condition H_v used here consists of the following inputs:

1. A computable upper bound for $D_\nu(c)$, obtained from enclosures at finitely many Laplace nodes, provable quadrature enclosures, and a bound on the infinite tail.
2. Outward enclosures for the finite set of transform loadings required in Section 3. For the discrete kernel, each loading also has a computable bound on the difference between the true kernel and its algebraic recursion.
3. Complete bounds on the infinite-frequency tail and periodization error in Fourier inversion, together with branch control for the continuous complex transform.

These inputs are defined in terms of specific transforms or moments. Their connection to the target price is established by the theorems below. We write $\mathsf{H} = \mathsf{H}_s + \mathsf{H}_v$. The finite catalogs, precision, tail-bound constants, and kernel error formulas are specified in Appendices A–C.

Theorem 4.1 (A signed enclosure of the price difference between two models). Suppose that P, Q_h each satisfy $\mathsf{H}_s(c)$, and that computable constants satisfy $D_P(c) \le d_P, D_Q(c) \le d_Q$. If H_v provides

$$e^{-rT}(E_Q\mathcal{L} - E_P\mathcal{L}) \in [\ell, u], \tag{4.1}$$

then, with $\gamma = e^{-rT}/(2\sqrt{2\pi})$,

$$\boxed{e_h \in \left[\ell - \gamma\left(\frac{d_Q}{K_2} + \frac{d_P}{K_1}\right),\quad u + \gamma\left(\frac{d_Q}{K_1} + \frac{d_P}{K_2}\right)\right].} \tag{4.2}$$

Proof. Apply (3.5) to each law and subtract. The lower-end contributions are $-R_{K_2,Q} - R_{K_1,P}$, and the upper-end contributions are $R_{K_1,Q} + R_{K_2,P}$. Lemma 3.1 and (4.1) give the result. □

Once the probabilistic and analytic bounds are in place, the theorem itself is interval combination; the mathematical substance lies in Lemma 3.1 and in the rigorous verification of H_v. The resulting interval is generally not symmetric about zero, and the computation keeps its center.

Corollary 4.2 (The specified Heston parameter point). For (2.1)–(2.5), take

$$\theta_* = (3,\ 9/200,\ 23/100,\ -11/20,\ 9/200), \qquad c = \frac{1254433}{1250000}. \tag{4.3}$$

Then

$$e_h \in [-.011024692273,\ .010642371599], \qquad |e_h| < .011025 < .025. \tag{4.4}$$

Furthermore,

$$\begin{aligned} p_c &\in [6.508371733,\ 6.518868974], \\ p_h &\in [6.507844281,\ 6.519014106]. \end{aligned} \tag{4.5}$$

Proof. Section 5 establishes the common Gaussian structure. Appendices A–C specify all requirements for the validated arithmetic computation, which yields

$$W_P := E_P[(A - cG)^2 I^{-1/2}] \le 2.172471592802, \qquad W_Q \le 2.242189951600. \tag{4.6}$$

Here $d_\nu = W_\nu/\sqrt{1-\rho^2}$. The center of the finite linear difference is approximately $-.000202992309237006$, and its complete error radius is at most $.000594238060$. Combining the exact rational endpoints outward with (4.2) gives (4.4). Applying the corresponding error contributions to (3.5) separately for each law gives (4.5). The source code, exact endpoints, and the recomputation of the entire catalog by a second implementation together constitute the executable evidence specified in the appendices. □

4.1 The optimal reference scale and the loss of cancellation

The common scale c can be chosen by a projection argument. The objective is the sum, under the two laws, of the squared arithmetic-reference residual weighted by inverse conditional standard deviation. This is a weighted least-squares problem with a single coefficient. Assume, in addition to the structural condition, that

$$m_{AA,\nu} = E_\nu[A^2/\sigma_\nu], \qquad m_{AG,\nu} = E_\nu[AG/\sigma_\nu], \qquad m_{GG,\nu} = E_\nu[G^2/\sigma_\nu]$$

are finite for $\nu = P, Q$. Put

$$\begin{aligned} a &= m_{AA,P} + m_{AA,Q}, \qquad b = m_{AG,P} + m_{AG,Q}, \\ d &= m_{GG,P} + m_{GG,Q} > 0. \end{aligned}$$

The nonlinear contribution to the width in (4.2), using exact moment inputs, is

$$\gamma\left(K_1^{-1} + K_2^{-1}\right) J(c), \qquad J(c) = a - 2bc + dc^2.$$

Consequently its unique minimizer on $c > 0$ is

$$c_* = \frac{b}{d}, \qquad J(c) = J(c_*) + d(c - c_*)^2. \tag{4.7}$$

The discount and strike factors are positive constants and therefore do not alter this minimizer. This statement concerns the exact nonlinear width; quadrature slack and the linear-transform error allowances can depend on c, so it does not assert that c_* minimizes the final computed enclosure.

In the Heston construction $\sigma_\nu = \sqrt{1-\rho^2}\sqrt{I_\nu}$, so the common factor $(1-\rho^2)^{-1/2}$ cancels from (4.7). A numerically preferable representation uses the arithmetic–geometric mean inequality:

$$H_\nu = E_\nu[G^2 I_\nu^{-1/2}], \qquad E_\nu^+ = E_\nu[(A-G)GI_\nu^{-1/2}] \geq 0, \qquad c_* = 1 + \frac{E_P^+ + E_Q^+}{H_P + H_Q}. \tag{4.8}$$

Both moments draw on profiles already present in the 91-profile catalog. The Laplace integrand $E_\nu[(A-G)G\exp(-x^2 I_\nu)]$, for example, is nonnegative and nonincreasing, so the same Darboux construction as (A.5) encloses it, with the difference of the AG and G^2 profiles evaluated at each node before integration. The positive AG tail bounds its continuous tail. At $S_0 = 100$, each positive raw-moment catalog has coefficient sum 10000 and the nodal projection allowance for $(A-G)G$ is $20000\delta_Q$. Working this way preserves cancellation at the nodal level and spares us from subtracting two wide integrated moment enclosures.

If validated positive endpoints give $E_P^+ + E_Q^+ \in [e_-, e_+]$ and $H_P + H_Q \in [d_-, d_+]$, with $d_- > 0$, then

$$c_* \in \left[1 + \frac{e_-}{d_+},\ 1 + \frac{e_+}{d_-}\right] = [c_-, c_+].$$

Choosing the rational midpoint $\hat{c} = (c_- + c_+)/2$ gives

$$0 \leq J(\hat{c}) - J(c_*) \leq \frac{d_+(c_+ - c_-)^2}{4\sqrt{1-\rho^2}}, \tag{4.9}$$

where d_+ encloses the sum of the two $I^{-1/2}$-weighted G^2 moments, as in (4.8). A ratio enclosure alone is not a new price certificate, because the selected rational scale still has to be substituted into the original square and linear calculations.

What optimizing c does not do is recover cancellation between the two models. Even with $Q = P$, an exact linear difference of zero, and exact $d_P = d_Q = D_P(c)$, formula (4.2) has width

$$2\gamma\left(K_1^{-1} + K_2^{-1}\right)D_P(c).$$

That width is positive unless $A = cG$ almost surely, so (4.2) is a fixed-step enclosure, and by itself it gives no proof that its width tends to zero as h decreases.

For computed inputs the actual width in (4.2) is

$$(u - \ell) + \gamma(d_P + d_Q)(K_1^{-1} + K_2^{-1}). \tag{4.10}$$

The original nonlinear interval has width approximately .02047859; its linear width is approximately .00118848, giving total width approximately .02166706. About 94.5 percent is therefore due to payoff conversion. Increasing precision or reducing an already small projection contribution has limited effect on the final width. Appendix H gives a sufficient coupling inequality for the difference of the remainders. Its additional moments have not been computed at the main point, so Corollary 4.2 retains its separate-law remainder. In the deterministic example of Appendix H, the nonlinear width falls from approximately

.01538363715 to .00008815020, about 0.573 percent of the separate-law width. Both calculations use the same conditioning and common geometric scale. This comparison concerns only that example's nonlinear component.

The positive-moment calculation at $h = 1/768$ gives

$$c_* \in [1.00342739460346610094 6724,\ 1.00366983109600091545 2372]. \tag{4.11}$$

Rounding a midpoint to the 10^{-7} grid selects the rational $5017743/5000000 = 1.0035486$, which is not the exact midpoint of the rational endpoints in (4.11). Formula (4.9) applies to an exact midpoint, so for the rounded choice the distance allowance also includes $5 \cdot 10^{-8}$.

Table 3. Effect of the common geometric scale on the nonlinear width.

Common scale	**Nonlinear width upper bound at $h = 1/768$**
1	.040551110717
1.0035464 (original)	.020478587752
1.0035486 (rounded midpoint)	.020478592980

The original scale is already close to the optimizer. The midpoint-based candidate slightly increases the computed upper width, so the original rational scale is retained. This comparison includes the new moment quadrature, projection, and tail allowances. It does not claim that the exact objective or the complete price certificate has been globally optimized.

5 How the models supply the common factor

5.1 The deterministic-variance benchmark

For the benchmark, the model's variance coefficient itself is a deterministic function $v(t) \geq 0$, the stock price follows the corresponding linear geometric diffusion, and

$$0 < I_1 = \int_0^{t_1} v(t)\,dt, \qquad \int_0^T v(t)\,dt < \infty. \tag{5.1}$$

Computability conclusions also require the integrated variances over all fixing intervals to be supplied as effectively computable inputs.

Take the stock Brownian integral over the first interval as U. Then $\sigma^2 = I_1$ is a positive constant; the remaining log increments and deterministic drift can all be included in $\mathcal{G}$. Every observed price has the common factor in (2.6). The finite positive moments of the joint lognormal distribution give

$$D_\nu(c) = I_1^{-1/2} E_\nu (A - cG)^2 < \infty.$$

The second moment is computed directly from

$$E(A - cG)^2 = \frac{1}{n^2} \sum_{i,j} E S_i S_j - \frac{2c}{n} \sum_i E S_i G + c^2 E G^2 \tag{5.2}$$

whose terms are all Gaussian exponential moments. The corresponding finite transforms and Gaussian tails likewise have explicit bounds. Thus, for a single law with effectively computable inputs, H holds; for two such laws, Theorem 4.1 gives the price-difference enclosure.

So the classical lognormal conditioning models fit within the framework. Once the common structure is identified, the same payoff-level argument applies to different variance mechanisms, and the particular constants and computational catalog come from the model itself.

5.2 Heston dynamics and the projected variance kernel

For the stochastic-variance calculation we use $\kappa, \bar{v}, \xi, v_0 > 0$, $|\rho| < 1$, the model (2.1) or (2.2), observation dates aligned with the discrete grid, and the finite-transform, moment-growth, projection, and infinite-tail inequalities listed in Appendices A–C. The actual verification in this section uses S_0, r, T, h from (2.5) and the parameters in (4.3); the common Gaussian construction below is valid under the structural parameter assumptions just stated.

Condition on the entire variance driver W and on the later independent stock-driver increments $B_t - B_{t_1}$, $t \geq t_1$. The conditioning sigma-field does not include the future Brownian levels B_t, which contain the first-interval value. Let

$$I_P = \int_0^{t_1} V_t \, dt, \qquad I_Q = h \sum_{j=0}^{t_1/h-1} V_j^h.$$

The contribution of B over the first interval is shared by every observation date. Hence

$$U \mid \mathcal{G} \sim N(0, (1-\rho^2) I_\nu), \qquad S_{t_i} = e^U a_i. \tag{5.3}$$

For the continuous model, $v_0 > 0$ and continuity of the variance paths imply $I_P > 0$ almost surely. For the discrete model, there is the exact lower bound

$$I_Q \geq h v_0 > 0. \tag{5.4}$$

The transform and tail bounds in the appendices verify $W_\nu < \infty$ and supply an upper bound, while the exponential-moment growth bounds ensure integrability of the required price terms. These inputs verify H for the two laws. At the parameters in (4.3), all these inequalities and finite-node checks hold.

5.3 Why a boundary atom does not destroy smoothing

The Gaussian factor depends on integrated variance over the first fixing interval and does not need the variance to have a positive density at every grid point. That matters for a projected scheme, whose one-step transition can have an atom at zero.

At the parameters in (4.3), the continuous variance is random:

$$\operatorname{Var}(V_t) = \xi^2 \int_0^t e^{-2\kappa(t-s)} E V_s \, ds > 0 \quad (t > 0), \tag{5.5}$$

so its variance is not a deterministic function. The first step of the discrete model also satisfies

$$Q_h(V_1^h = 0) = \Phi\left(-\frac{(1-\kappa h)v_0 + \kappa \bar{v} h}{\xi \sqrt{h v_0}} \right) \in (0, 1). \tag{5.6}$$

At the same time, (5.4) still holds. The positive initial variance makes the first accumulated variance positive even on paths that later reach the projected boundary. The common factor therefore survives the atom. The atom still changes the transform law, and Appendix B must account for that change; conditioning does not remove the projection error.

6 An exact Gaussian benchmark for weak expansions

A fixed-step enclosure and a weak-error expansion answer different questions. The former bounds the price difference at a chosen grid; the latter identifies its leading response as the step tends to zero. To derive the expansion directly, we now use the exactly solvable variance dynamics on the deterministic-variance face. For background on weak error expansions and distributional error analysis, see [8, 9, 10]. This section works directly with the differentiability of Gaussian densities. We set $\xi = 0$, retain the original update (2.2), and assume

$$\kappa > 0, \quad \bar{v}, v_0 > 0, \quad t_j = j/12, \quad h = 1/(12m) \le h_0 < 1/\kappa, \quad m \in \mathbb{N}. \tag{6.1}$$

Let $D = v_0 - \bar{v}$. The continuous and discrete cumulative variances are

$$I(t) = \bar{v}t + \frac{D}{\kappa}(1 - e^{-\kappa t}), \qquad I_h(t) = \bar{v}t + \frac{D}{\kappa}[1 - (1 - \kappa h)^{t/h}]. \tag{6.2}$$

The twelve independent components of the log-increment vector Y have distributions

$$N(r/12 - s_j/2, s_j), \qquad s_j = I(t_j) - I(t_{j-1}), \tag{6.3}$$

and the discrete law replaces s_j by $s_{h,j}$. Both satisfy the lower bound $s_j, s_{h,j} \ge u_* := \min(v_0, \bar{v})/12 > 0$.

Define

$$a(t) = \frac{D\kappa t e^{-\kappa t}}{2}, \quad B_I(t) = |D| e^{-\kappa t}\left[\frac{\kappa^2 t}{3(1-\kappa h_0)} + \frac{\kappa^3 t^2}{8(1-\kappa h_0)^2}\right]. \tag{6.4}$$

Set $d_j = a(t_j) - a(t_{j-1})$ and $b_j = B_I(t_j) + B_I(t_{j-1})$. Appendix D proves that

$$|s_{h,j} - s_j - h d_j| \le h^2 b_j. \tag{6.5}$$

Theorem 6.1 (A common first-order expansion for bounded payoffs). Under (6.1), let $g(Y)$ be any bounded Borel payoff with range contained in $[l, l + L]$, payable at time τ. Then

$$p_h(g) - p_c(g) = h\beta_g + R_h(g), \tag{6.6}$$

$$\beta_g = e^{-r\tau} E_s\left[g(Y) \sum_j d_j \mathcal{S}_j(Y)\right], \tag{6.7}$$

$$|R_h(g)| \le e^{-r\tau}\frac{L}{2} C_2 h^2,$$

where

$$\mathcal{S}_j = \frac{G_j^2 - 1}{2s_j} - \frac{G_j}{2\sqrt{s_j}}, \quad J(s) = \frac{1}{2s^2} + \frac{1}{4s}, \quad H_2(s) = \frac{2}{s^2} + \frac{1}{2s},$$

$$C_2 = \sqrt{\sum_j b_j^2 J(u_*)} + \frac{1}{2}\sum_j (|d_j| + h_0 b_j)^2 H_2(u_*). \tag{6.8}$$

Here G_j denotes the standardized independent normal increments.

Proof. The Gaussian density is twice differentiable in L^1 with respect to its variance parameters. The norm of its first directional derivative is bounded by the corresponding square root of the Fisher information quadratic form, while its second directional derivative satisfies

$$\|Dp_s[z]\|_1 \le \sqrt{\sum_j z_j^2 J(s_j)}, \qquad \|D^2 p_s[z,z]\|_1 \le \sum_j z_j^2 H_2(s_j).$$

Substituting (6.5) into the L^1 Taylor formula for the density gives $\|p_{s_h} - p_s - hDp_s[d]\|_1 \le C_2 h^2$. The density remainder integrates to zero, so the payoff can be centered as $g - l - L/2$, yielding (6.7). Appendix D gives the derivative calculations and the full justification by dominated convergence. □

The theorem applies the same leading density perturbation to all ten payoffs. For a put,

$$\beta_{K,\tau} = a(\tau)\frac{S_0\varphi(d_1)}{2\sqrt{I(\tau)}}, \quad d_1 = \frac{\log(S_0/K) + r\tau + I(\tau)/2}{\sqrt{I(\tau)}}. \tag{6.9}$$

When $v_0 \neq \bar{v}$, all nine put components are nonzero and have the same sign as $v_0 - \bar{v}$.

On the family

$$\kappa = 3, \quad \bar{v} = .045, \quad v_0 \in [.03, .06], \quad h_0 = 1/768, \tag{6.10}$$

the second-order remainder for the original Asian payoff has the uniform upper bound .000165217. A separate finite-step density comparison also gives

$$|p_h(\psi) - p_c(\psi)| \le .004990924. \tag{6.11}$$

The leading Asian coefficient has the exact integral representation (6.7) and an effective absolute bound. This common expansion handles several nonsmooth payoffs through one integrable density perturbation, and every remainder constant stays explicit.

6.1 Asian coefficient bounds and numerical illustration

The score bound from (6.7) and the first derivative norm gives

$$|\beta_{\text{Asian}}| \le \frac{15e^{-rT}}{2}\sqrt{\sum_{j=1}^{12} d_j^2 J(s_j)}. \tag{6.12}$$

Table 4 gives pointwise coefficient and remainder bounds for the numerical family (6.10), with the remainder bounds coming from (6.5)–(6.8) and each point's lower increment variance. This gives a common expansion formula and explicit remainder bounds. The coefficient enclosures contain zero, and the signed twelve-dimensional Asian coefficient integral remains to be evaluated.

Table 4. Analytic Asian coefficient enclosures and remainder bounds.

Initial variance	Certified coefficient enclosure	Upper bound on $\lvert e_h - h\beta\rvert$, $h = 1/768$
0.03	$[-3.441491358, 3.441491358]$	0.000121667773
0.04	$[-0.928189505, 0.928189505]$	0.000036646884
0.05	$[-0.785853082, 0.785853082]$	0.000034532369
0.06	$[-2.057270822, 2.057270822]$	0.000099684585

For a Monte Carlo illustration we use 2,097,152 Gaussian increment vectors per point, with common samples across the four step sizes in Table 5. In increasing order of initial variance, the coefficient estimates are -0.041015 (SE 0.002079), -0.011364 (SE 0.000566), 0.008329 (SE 0.000483), and 0.019917 (SE 0.001274). They are sampling estimates and should be kept distinct from the analytic enclosures in Table 4.

Table 5. Sampling estimates of the scaled Asian expansion residual.

Initial variance	$h = 1/192$	$1/384$	$1/768$	$1/1536$
0.03	-0.063243	-0.062944	-0.062796	-0.062722
0.04	-0.017226	-0.017145	-0.017105	-0.017085
0.05	0.011698	0.011646	0.011620	0.011607
0.06	0.027912	0.027785	0.027722	0.027690

The entries estimate $(e_h - h\beta_{\mathrm{Asian}})/h^2$ through a Gaussian density-ratio calculation. At $h = 1/768$ their estimated standard errors are .003733, .001000, .000863, and .002319, respectively, and estimates across step sizes at a point are correlated. The scales stay stable, which is consistent with the second-order expansion. The $h = 1/192$ and $1/384$ columns are diagnostics, since they lie outside the fixed $h_0 = 1/768$ range used for the numerical constants in (6.10) and Table 4.

7 What the forward bounds imply after calibration

A numerical change affects the prices entering a calibration likelihood as well as the target price evaluated after calibration. The fixed-parameter target-price certificate alone does not control the change in the posterior. This section adds an explicit transport bound for the parameter posterior, so that both changes enter the target-quantile comparison. The calculation uses a restricted near-deterministic family and synthetic quotes; it does not establish calibration stability near the main stochastic-variance example.

7.1 Price transfer for small positive volatility of volatility

Suppose that the deterministic reference variance and its Euler values lie in $[m, M]$, where $0 < m \le M < \infty$ and $\kappa \ge \kappa_- > 0$, and assume

$$\kappa h \le 1, \qquad 0 \le \xi \le \sqrt{\kappa_- M/2}. \tag{7.1}$$

Define

$$C_P = \sqrt{M/(2\kappa_-)}, \qquad C_Q = \sqrt{2M/\kappa_-}.$$

The Itô isometry for the continuous model and the 1-Lipschitz property of the discrete positive-part projection give, respectively,

$$E(V_t^{\xi} - v(t))^2 \le \xi^2 C_P^2, \qquad E(V_j^{\xi,h} - v_j)^2 \le \xi^2 C_Q^2. \tag{7.2}$$

Combining the deterministic reference lower bound m with Doob's inequality yields

$$E\max|Z_\ell^\xi - Z_\ell^0| \le \xi C_\ell\left(\frac{T}{2} + 2\sqrt{T/m}\right), \quad \ell = P, Q. \tag{7.3}$$

In log stock price the original put payoff is K-Lipschitz, and in the maximum norm of the fixing log-price vector the original spread payoff is $K_2 = 110$-Lipschitz. Thus (7.3) provides uniform radii from all true calibration prices, and from the true target price, to their reference-family counterparts. Appendix E gives the inductive proof, which retains the effect of projection on the mean.

Take, for example, the parameter box of positive five-dimensional volume

$$\begin{gathered}\kappa \in [2.999, 3.001], \quad \bar v \in [.04499, .04501],\\ v_0 \in [.03001, .05999], \quad \xi \in [10^{-5}, 4\cdot 10^{-5}], \quad \rho \in [-.8, -.3],\end{gathered} \tag{7.4}$$

where the parameter coordinates are ordered as $(\kappa, \bar v, \xi, \rho, v_0)$. On this box, finite verification gives an absolute bias bound of .020948 for the original Asian and .016702 for each of the nine puts.

7.2 The full joint posterior

Forward numerical error acts on the likelihood and on its normalizing constant, a basic issue in Bayesian approximation [11]. In this section, set $\kappa = 3, \bar v = .045$, and let

$$u = v_0 \sim U[.03, .06], \quad \xi \sim U[10^{-7}, 10^{-6}], \quad \rho \sim U[-.8, -.3] \tag{7.5}$$

be mutually independent. Write $\zeta = (\xi, \rho)$ and denote its prior probability measure by ν. Throughout this posterior calculation and Appendix F, $m \in \{P, Q\}$ indexes the pricing model; ν remains the nuisance-parameter prior. For model $m = P, Q$, let $c_m(u, \zeta)$ be the true calibration vector and $J_m(u, \zeta)$ the true Asian price. Given synthetic quotes y and a positive-definite covariance matrix Σ_d, define

$$w_m(u,\zeta) = \exp\left[-\frac{1}{2}(c_m - y)^T\Sigma_d^{-1}(c_m - y)\right], \quad \bar w_m(u) = \int w_m(u,\zeta)\nu(d\zeta),$$

$$Z_m = \frac{1}{.03}\int_{.03}^{.06} \bar w_m(u)\,du, \qquad \mu_m(du, d\zeta) = \frac{w_m(u,\zeta)}{.03 Z_m}\,du\,\nu(d\zeta). \tag{7.6}$$

Each model uses its own calibration prices, normalizing constant, and target price.

7.3 A transfer theorem for every quantile level

Theorem 7.1. Suppose that the two marginal posteriors of u admit a coupling (U_P, U_Q) such that $|U_P - U_Q| \le \delta_u$ almost surely. Assume that the reference targets satisfy

$$|J_P^0(u) - J_P^0(v)| \le L_0|u - v|, \quad |J_Q^0(u) - J_P^0(u)| \le \varepsilon_0,$$

and that the true targets satisfy $|J_m(u,\zeta) - J_m^0(u)| \le e_{m,J}$. Then the left generalized quantiles of the true targets under their respective full joint posteriors satisfy

$$\boxed{|q_Q(p) - q_P(p)| \le \varepsilon_0 + e_{P,J} + e_{Q,J} + L_0\delta_u \quad (0 < p < 1).} \tag{7.7}$$

Proof. For each model, define the conditional probability kernel

$$K_m(u, d\zeta) = \frac{w_m(u,\zeta)}{\bar w_m(u)}\nu(d\zeta).$$

Extend the marginal coupling above by drawing ζ_P and ζ_Q from $K_P(U_P, \cdot)$ and $K_Q(U_Q, \cdot)$, respectively. Fubini's theorem shows that the resulting full marginals are exactly μ_P and μ_Q. The triangle inequality gives

$$|J_Q(U_Q, \zeta_Q) - J_P(U_P, \zeta_P)| \le \varepsilon_0 + e_{P,J} + e_{Q,J} + L_0\delta_u =: E.$$

Consequently, $F_P(z - E) \le F_Q(z) \le F_P(z + E)$. Taking left generalized inverses yields (7.7). The argument also applies to target distributions with atoms. □

7.4 Effective inputs and numerical conclusions

We partition $[.03, .06]$ into 4096 equal intervals and, on each interval and over the entire ζ-domain, enclose the true likelihood using (7.3) and the reference Black prices. The same bounds remain valid after integration against the normalized prior ν, so we can construct enclosures of the marginal CDFs while retaining their shared denominators, and an inverse-CDF coupling driven by a common uniform variable then gives δ_u. Appendix F gives all constants and the finite pairing algorithm.

The single-quote case uses the put with $T = 1, K = 100$, the nine-quote case uses the full calibration vector, and both use $\Sigma_d = .75^2(.75I_d + .25\mathbf{1}\mathbf{1}^T)$. Appendix F lists the exact quotes. The results are

$$L_0 \le 213.792201829985, \quad \varepsilon_0 \le .004493504474, \quad e_{P,J} + e_{Q,J} \le .000393595478,$$

Case	δ_u	$\sup_{0<p<1} \lvert q_Q(p) - q_P(p)\rvert$
Single quote	$3/204800$	$\le .008018821658$
Nine quotes	$3/81920$	$\le .012716404217$

(7.8)

The finite partition encloses the continuous posterior and retains the full normalizing mass of each model.

8 Numerical consequences of the error decomposition

The numerical calculations take up three questions raised by the preceding analysis: whether the separate-law bound meets a fixed monetary tolerance, which term keeps the width from shrinking under refinement, and whether a coupling can reduce that term in an analytically tractable case. The weak-expansion and posterior calculations keep their separate domains from Table 2. Simulation diagnostics are labeled as such and are not used to certify any interval.

8.1 From formulas to certified monetary bounds

Table 6 assembles the price difference at the original parameter point, with displayed enclosures and upper bounds rounded outward. The exact endpoints, numerical constructions and reproduction instructions are in Appendix G.

Table 6. Error budget for the principal Heston price certificate.

Quantity	Validated upper bound or enclosure
Nonlinear conversion interval	$[-.010227461904, .010251125849]$
Q projection error per linear profile	.000000012711979137
Q projection contribution over all linear profiles	.000004310413
Omitted frequency tail for the continuous model	.000014095712
Omitted frequency tail for the discrete model	.000184373301
Periodization error for one model	.000195729318
Total error bound for the linear difference between the two models	.000594238060
Absolute bound on the true arithmetic Asian bias	**.011025**

We compute reported widths from the exact ledger and round them outward, so subtracting displayed endpoints can differ in the final decimal place. For $[\ell, u]$, width means $u - \ell$, midpoint radius means $(u - \ell)/2$, and the absolute bias bound is $\max(|\ell|, |u|)$.

8.2 Step-grid certificates and the width plateau

Table 7 changes the Euler step and retains the main Heston parameters, payoff, fixing dates and transform catalogs. Every successful row includes the full error accounting and the alternative numerical checks described in Appendix G.

Table 7. Signed Heston price-difference certificates at four Euler grids.

Step	Euler minus continuous enclosure	Width upper bound
1/192	No certificate: projection guard fails	Not available
1/384	$[-0.014169989568, 0.013839530031]$	0.028009519599
1/768	$[-0.011024692273, 0.010642371599]$	0.021667063872
1/1536	$[-0.010967952032, 0.010766337995]$	0.021734290027

The first row fails because of the fixed projection proof: at $h = 1/192$ it gives $\beta = 184040303/23040000 < 8$, while its recurrence requires $\beta > 8$, so this sufficient condition

does not certify that mesh. All three successful rows satisfy the absolute error tolerance .025. Table 8 separates their width contributions, and its exact unrounded entries sum to the interval widths.

Table 8. Width contributions in price units for the three successful grids.

Width contribution	1/384	1/768	1/1536
Payoff remainder	0.026754977094	0.020478587752	0.020171631177
Linear projection	0.000265787793	0.000008620825	2.65628E-7
P frequency tail	0.000028191423	0.000028191423	0.000028191423
Q frequency tail	0.000177646019	0.000368746602	0.000751284529
Periodization	0.000782917272	0.000782917272	0.000782917272
Rounding and endpoints	1E-12	1E-12	1E-12

The entries of Table 8 are width contributions, not one-sided error radii, and rounding them separately can change their displayed sum. As the grid is refined the nonlinear shares are approximately 95.5, 94.5, and 92.8 percent. The total width no longer improves at the finest step, where the discrete frequency-tail contribution grows by more than the payoff remainder and projection contribution shrink. The frequency-tail bound in (C.7) uses the first-step lower bound $i_0 = hv_0$, so refining the Euler grid alone does not tighten this fixed inversion design. The plateau describes the enclosure, not the true price error.

8.3 A second stochastic-variance parameter point

Keep the contract, twelve monthly fixings, $h = 1/768$, and all parameters in (4.3) except the initial variance, which is now $v_0 = .04$. We evaluate the weighted moments, projection prefixes, transforms, and continuous and discrete tails afresh at this point. The same growth estimate gives $\exp(.6 + 4v_0) = \exp(.76)$, so the retained $\exp(.78)$ moment allowance remains conservative. All real-loading, denominator, coefficient and complex branch conditions pass both numerical constructions.

Table 9. Complete Heston certificate at initial variance .04.

Quantity	**Certified enclosure or upper bound**
Continuous price	$[6.499318797594, 6.509948941256]$
Projected-Euler price	$[6.498150434416, 6.510702536853]$
Euler minus continuous	$[-0.011798506840, 0.011383739259]$
Signed-interval width	0.023182246099

The enclosure combines the nonlinear remainder with projection, both frequency tails, periodization and outward arithmetic, and Appendix G's package records its six exact width contributions and the complete alternative checks. Relative to $v_0 = .045$ the signed-interval width increases by about 6.99 percent, and about 83.2 percent of that increase comes from the discrete frequency-tail contribution, which rises from approximately .0003687466014 to

.0016298806683. The tail bound weakens for two reasons visible in (C.7): the smaller v_0 lowers the first-step floor $i_0 = hv_0$, and the negative first-month Laplace coefficient also increases $\overline{L}_Q(U) = \exp(A + bv_0)$. The comparison measures enclosure tightness, not an increase in the true bias. This is a second pointwise certificate with the same Feller index.

8.4 Coverage and sensitivity design

The main point has Feller index $2\kappa\bar{v}/\xi^2 = 2700/529$, approximately 5.104. Testing sensitivity to the variance boundary, correlation, strikes, initial variance and fixing frequency takes new moment, projection and tail inputs for each case. Changing to 52 or 252 fixings, for instance, requires an aligned Euler grid, so $h = 1/768$ cannot be retained, and the first fixing also changes the weight $I_1^{-1/2}$ in the payoff bound. The path comparisons below explore these sensitivities and leave the certified parameter domain where it was.

8.5 Independent path diagnostics for boundary and fixing sensitivity

The path study couples an Euler grid to one with twice as many steps, using 32,768 paths per case. Table 10 reports three of the fourteen cases, selected by Feller index; the full study also varies correlation, strikes, initial variance and step size. The coarse-minus-fine estimates have sampling error and retain the finer model's discretization bias.

Table 10. Coupled path diagnostics at three Feller indices.

Case	Coarse minus fine, estimate and standard error	Projection fraction, coarse
Feller 5.104	-0.0000269; SE 0.0004717	0.0000000
Feller 0.999	0.0001129; SE 0.0012453	0.0011268
Feller 0.639	0.0009133; SE 0.0018375	0.0066991

All three price-difference estimates have magnitude below one estimated standard error, so Table 10 resolves neither their signs nor a systematic increase in pricing bias. The proposal frequencies support only the increased incidence of negative variance proposals near and below the boundary; they are not projection-error bounds.

The fixing comparison shows the sensitivity of the smoothing weight, though it also changes the time grid. For 12, 52, and 252 fixings, the sampled means of $I_1^{-1/2}$ are 16.489, 34.078, and 74.871, on aligned coarse grids of 768, 832, and 1008 steps, respectively, with twice as many fine steps.

8.6 Continuous-model valuation and numerical risk budgets

Suppose that a valuation system provides a price enclosure $[q_L, q_U]$ for the same discrete model. Theorem 4.1 gives $e_h \in [\ell, u]$, hence the continuous-model price satisfies

$$p_c \in [q_L - u,\ q_U - \ell]. \tag{8.1}$$

For (4.3), the direct continuous-price interval (4.5) has width below .010498. These two routes give enclosures in price units for valuation and implementation checks.

8.7 Stability of the price distribution after calibration

Changing the numerical implementation affects both the calibration likelihood and the target price, and Theorem 7.1 retains both effects: $\varepsilon_0 + e_{P,J} + e_{Q,J}$ controls the target-price difference at fixed parameters, and $L_0\delta_u$ controls the target change caused by movement of the parameter posterior.

On the original 4096-cell grid, the transport contribution accounts for about 39 percent of the single-quote bound and 62 percent of the nine-quote bound, which is the reason for the grid refinement in the next subsection. The nine-quote enclosure is wider because of its conservative transport contribution, and that does not imply a larger true numerical bias.

8.8 Posterior grid refinement

Table 11 keeps the same prior (7.5), synthetic observations, likelihood covariance and target Lipschitz constant. Each row verifies the full-cell likelihood bounds and coupling inequalities and bounds the displacement of every Asian-price quantile.

Table 11. Certified Asian-price quantile displacement under posterior grid refinement.

Cells	Single quote	Nine quotes	Coupling steps, single/nine
4096	0.008018821658	0.012716404217	2 / 5
8192	0.006452960804	0.010367612938	2 / 7
16384	0.006061495591	0.009584682511	3 / 12

The common nonmesh error contributions remain unchanged. Coupling step counts increase with refinement, so the final bound does not scale exactly as the cell width. The new bounds at 16384 cells are approximately 24.4 percent and 24.6 percent smaller than the original single- and nine-quote bounds. Adaptive cells and local target Lipschitz constants are not evaluated here.

The absolute 2.5, 50, and 97.5 percent quantile brackets concern the initial-variance marginal. For the continuous-model posterior with 4096 cells, the single-quote brackets are approximately [.0307690,.0307764], [.0444067,.0444141], and [.0590917,.0590992]; the nine-quote brackets are [.0308203,.0308350], [.0435131,.0435425], and [.0587182,.0587256]. These brackets describe parameter uncertainty. Absolute Asian-price target quantiles have not been evaluated, so the displacement bounds do not yet measure the change relative to the target's posterior uncertainty scale.

9 Domain of validity

The structural conditions are sufficient for certification. Section 5 verifies a deterministic-variance class and the specified Heston model and projected kernel, and Section 8 varies the step at the principal point and certifies a second initial variance. Extending these enclosures to a parameter neighborhood requires uniform moment, transform, projection and tail inequalities. The small-positive-ξ box in Section 7 is a separate certified region.

Section 6 gives a common expansion formula and explicit remainder bounds for $\xi = 0$. The nine put coefficients are nonzero when $v_0 \neq \bar{v}$, whereas for the Asian coefficient we have a score integral and analytic bounds containing zero. An expansion for positive ξ requires additional regularity and integrable remainders for the projection kernel and payoff kinks.

The posterior displacement bounds apply to the restricted prior (7.5) and synthetic quotes. Transfer on the five-dimensional parameter domain

$$[2,4]\times[.03,.06]\times[.18,.28]\times[-.8,-.3]\times[.03,.06]$$

in coordinate order $(\kappa,\bar v,\xi,\rho,v_0)$ requires uniform forward-error and posterior-transport inputs. Absolute Asian-price target quantiles and the signed Asian leading-coefficient integral remain to be computed.

The separate-law remainder produces the width plateau in Section 8. Proposition H.1 provides a direct coupled-remainder inequality. Its additional coupled moments are certified for a deterministic-variance example in Appendix H, but have not been established for the stochastic-variance Heston examples. Controlling those moments and tightening the discrete frequency tail are distinct mathematical requirements for a shrinking certificate width.

10 Conclusion

For the stated Heston model and projected Euler kernel, the paper gives a signed finite-step price enclosure with absolute bound below .011025, together with a validated continuous-price interval. The enclosure is built from three objects in sequence: an exact conditional lognormal price, a geometric reference compatible with the fixing-date transforms, and a convex remainder weighted by the available conditional smoothing. Together they determine the moment catalog and the error budget.

The same decomposition shows where the result stops. Separate bounds discard cancellation between the two remainder expectations and need not shrink when the laws approach one another. Refining a grid cannot by itself recover that information. The direct coupled estimate addresses the missing comparison and is evaluated here on the deterministic-variance face. Extending it to the stochastic-variance examples requires additional coupled-moment estimates. The weak-expansion and posterior-transfer results add consequences on their specified families and leave this boundary of the main certificate unchanged.

Appendix A. Finite certification of the weighted second moment

This appendix and Appendices B and C use the rational parameters in (4.3). Write

$$\alpha=\xi^2/2,\quad d=\kappa\bar v,\quad s=1-\rho^2,\quad t_1=1/12.$$

Throughout, c denotes the geometric scaling coefficient in (4.3).

A.1 A nonnegative integral and the loading catalog

For $I>0$,

$$I^{-1/2}=\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-x^2I}\,dx.$$

Applying Tonelli's theorem to the nonnegative square gives

$$W_\nu=\frac{2}{\sqrt{\pi}}\int_0^\infty F_\nu(x)\,dx,\qquad F_\nu(x)=E_\nu[(A-cG)^2e^{-x^2I}]. \tag{A.1}$$

We have $F_\nu \ge 0$, and the function is nonincreasing. In the expansion (5.2), the 144 ordered terms in A^2 combine into 78 terms; there are also 12 terms of the form $S_i G$ and one G^2 term. Each normalized exponential profile has total real stock loading 2, and the remaining loading in the backward monthly recursion satisfies $p \in [0, 2]$. The sum of the absolute coefficients of the 91 terms is

$$S_0^2(1+c)^2 = 10000(1+c)^2. \tag{A.2}$$

A killing rate x^2 is imposed in the first month and is zero in all other months.

A.2 The true continuous flow and the discrete algebraic recursion

On an interval with stock loading q and killing rate λ, write the normalized transform as $\exp(\mathcal{A} + \mathcal{B}v)$. The continuous coefficients satisfy

$$\mathcal{B}' = \alpha\mathcal{B}^2 - (\kappa - \rho\xi q)\mathcal{B} + (q^2 - q)/2 - \lambda, \qquad \mathcal{A}' = rq + d\mathcal{B}. \tag{A.3}$$

Completing the square in the one-step Gaussian integral gives the discrete algebraic recursion

$$\begin{aligned} \mathcal{A}_j &= \mathcal{A}_f + h(rq + d\mathcal{B}_f), \\ \mathcal{B}_j &= \mathcal{B}_f + h\left[\alpha\mathcal{B}_f^2 - (\kappa - \rho\xi q)\mathcal{B}_f + (q^2 - q)/2 - \lambda\right]. \end{aligned} \tag{A.4}$$

Both updates use the value $\mathcal{B}_f$ before the update. Appendix B bounds the difference between the actual positive-part kernel and (A.4). Fixed observation dates are handled by adding the corresponding stock loadings in reverse chronological order.

For real loadings $p \in [0, 2]$ and $\lambda \ge 0$, the continuous vector field is strictly downward at $\mathcal{B} = 1$. Thus the backward flow from terminal value 0 satisfies $\mathcal{B} \le 1$, and $\mathcal{A} \le (2r + d)T$. The uniform integrability established in Appendix B.3 identifies this flow with the true transform.

A.3 Validated Darboux enclosures

Take $\Delta x = 1/2$. For the continuous model, use nodes $x_j = j/2$, $0 \le j \le 256$; for the discrete model, use $0 \le j \le 128$. If the true nodal value lies in $[f_j^-, f_j^+]$, then

$$\Delta x \sum_{j=1}^{M} \max(0, f_j^-) \le \int_0^{M\Delta x} F_\nu(x)\,dx \le \Delta x \sum_{j=0}^{M-1} \max(0, f_j^+). \tag{A.5}$$

These are the rectangular-sum bounds for a nonnegative, nonincreasing function. For Q, first multiply (A.2) by the per-profile error bound from Appendix B and enlarge the algebraic nodal enclosure in both directions. Then apply (A.5).

A.4 The discrete infinite tail

Since $I_Q \ge i_0 = hv_0$, for $x \ge X$,

$$F_Q(x) \le F_Q(X) e^{-i_0(x^2 - X^2)}.$$

Using $x^2 - X^2 \ge 2X(x - X)$ gives

$$\int_X^\infty F_Q(x)\,dx \le \frac{\overline{F}_Q(X)}{2Xi_0}, \qquad X = 64. \tag{A.6}$$

The upper bound $\overline{F}_Q(X)$ includes the actual projection error and rounding.

A.5 The continuous infinite tail

Let $\kappa' = \kappa - 2\rho\xi$, and define

$$\Delta(x) = \sqrt{\kappa'^2 + 4\alpha(x^2-1)}, \quad r_1 = \frac{\kappa' - \Delta}{2\alpha}, \quad r_2 = \frac{\kappa' + \Delta}{2\alpha}.$$

The bound $\mathcal{B} \le 1$ in subsequent months allows each positive-loading profile to be bounded above by taking terminal value 1 for the first-month flow. For $x \ge X = 128$,

$$g = \frac{1 - r_1}{1 - r_2} \in (-1, 0).$$

The cross-ratio solution of the Riccati equation yields

$$\mathcal{B}(t_1) \le r_1 + \frac{\Delta}{\alpha} e^{-\Delta t_1}, \qquad \int_0^{t_1} \mathcal{B}(t)\,dt \le r_1 t_1 + \frac{\log 2}{\alpha}. \tag{A.7}$$

Furthermore, $\Delta(x) \ge 2\sqrt{\alpha}(x - 1/X)$ and $\Delta(X)t_1 > 1$, so $\Delta e^{-\Delta t_1}$ is decreasing throughout this tail region. Define

$$C_0 = \frac{\kappa'}{2\alpha} + \frac{1}{X\sqrt{\alpha}}, \quad C_1 = C_0 + \frac{\Delta(X) e^{-\Delta(X)t_1}}{\alpha},$$

$$C = \exp\!\left\{2r + d + 2rt_1 + d\left(t_1 C_0 + \frac{\log 2}{\alpha}\right) + v_0 C_1\right\}, \qquad \zeta = \frac{dt_1 + v_0}{\sqrt{\alpha}}.$$

Each normalized real transform is therefore at most $Ce^{-\zeta x}$. Since $(A - cG)^2 \le 2A^2 + 2c^2G^2$,

$$\int_X^\infty F_P(x)\,dx \le \frac{20000(1 + c^2)C}{\zeta} e^{-\zeta X}. \tag{A.8}$$

All root relations, signs, and transcendental functions are checked using outward-rounded arithmetic.

Combining (A.5)–(A.8) and multiplying by $2/\sqrt{\pi}$ gives

$$\begin{aligned} W_P &\le 2.17247159280158941544035 2, \\ W_Q &\le 2.24218995159904964746854 3. \end{aligned} \tag{A.9}$$

These moment bounds also establish the integrability required in (2.7).

Appendix B. The actual projection kernel and affine identification

B.1 The one-step projection correction

Let the current variance be $v \ge 0$, and let p be the real part of the stock loading. After the Gaussian tilt induced by the stock loading, the candidate next-step variance satisfies

$$Y \sim N(\eta_p v + dh,\ \xi^2 h v), \qquad \eta_p = 1 - (\kappa - \rho\xi p)h,$$

with the external factor $\exp(rph + \gamma_p h v)$, where $\gamma_p = (p^2 - p)/2$. For a future complex variance coefficient b, if the candidate is $Y = -z < 0$, then

$$|e^{bY^+} - e^{bY}| \le |b| z e^{|b|z} \le \frac{|b|h}{\lambda e} \exp[(\lambda/h + |b|)z]. \tag{B.1}$$

Set $s_b = \lambda + h|b|$. The Gaussian moment formula reduces the exponent in the state-dependent error bound to

$$-\frac{v}{h}\left(s_b\eta_p - \alpha s_b^2 - \gamma_p h^2\right) - s_b d. \tag{B.2}$$

Take $\lambda = 12$. Suppose p lies in the specified loading range, $\eta_p \geq \eta_*$, $\gamma_p \leq \gamma_*$, and $|b| \leq b_*$. Checking lower bounds for the concave quadratic at the two endpoints $s_b = 12$ and $s_b = 12 + b_* h$ suffices to ensure that the expression in parentheses is at least 8.

B.2 From local error bounds to true prefixes

Define

$$u_0 = 8, \qquad u_{j+1} = \eta_* u_j - \alpha u_j^2 - \gamma_* h^2. \tag{B.3}$$

Nonnegativity $u_j \geq 0$ is checked at all 768 levels. For $u \geq 0$, $e^{-uY^+/h} \leq e^{-uY/h}$. Hence the actual stock-weighted prefix Laplace expectations can be bounded above by repeated application of Gaussian moment bounds. This yields the projection error bound for each complete profile:

$$\delta_Q = \frac{b_*}{12e}\exp(C_f + C_{\mathrm{pre}} - 12d)\, h \sum_{j=0}^{767} \exp\left[-d\sum_{k<j} u_k - \frac{v_0 u_j}{h}\right]. \tag{B.4}$$

Here C_f bounds the future constant term and C_{pre} bounds the stock drift over the true prefix. The two parameter sets are as follows.

Table B.1. Constants for the projection error allowance.

Use	Range of p	b_*	η_*	γ_*	$C_f + C_{\mathrm{pre}}$
Weighted second moment	$[0, 2]$	512	$1 - (\kappa - 2\rho\xi)h$	1	$4r + d$
Thirteen linear transforms	$[-1/2, 1]$	1024	$1 - (\kappa - \rho\xi)h$	$3/8$	$2r + d$

The finite catalog checks both $\operatorname{Re} b \leq 1$ and the modulus bound for the future coefficients. In the first case, (B.4) is multiplied by $10000(1 + c)^2$ to obtain the error allowance at each Laplace node. In the second case, it is multiplied by the sum of the absolute values of all Fourier coefficients. No additional factor of h is applied to this sum.

B.3 Moment growth and removal of stopping

For real stock loading p, take e^{4V} as the variance component of the test function. The coefficient of V in the continuous generator calculation is

$$16\alpha - 4\kappa + 4\rho\xi p + \gamma_p. \tag{B.5}$$

This coefficient is strictly negative for all required loadings, including their $5/4$-scaled range $p \in [-5/4, 5/2]$. For Q, the positive coefficient 4 allows the direct use of $1 - e^{-4z} \leq 4z$. Together with the Gaussian tail bound at $\lambda = 12$, the additional growth rate caused by projection is at most

$$\frac{4e^{-12d}}{12e}.$$

Over the same enlarged range, the state coefficient in the tilted exponent remains nonpositive. For example, using the worst-case values $\eta_{5/2}$ and $\gamma_* = 15/8$, exact evaluation gives

$$12\eta_{5/2} - 144\alpha - \frac{15}{8}h^2 = \frac{8001336523}{983040000} > 0.$$

The total constant growth satisfies

$$4d + \frac{5}{2}r + \frac{4e^{-12d}}{12e} < .589267620943 < .6. \tag{B.6}$$

Thus the required exponential moments of the true continuous and discrete models admit uniform finite upper bounds. In particular, the normalized moments for the original loading range $p \in [-1, 2]$ are bounded by

$$M_2 = e^{4v_0 + .6} = e^{.78} \tag{B.7}$$

First stop the continuous affine expression on bounded variance domains. Raising its modulus to the power $5/4$ scales the stock loadings into the range above. The future positive variance coefficient is at most $5/4 < 4$, and the nonpositive killing contribution can be omitted in an upper bound. Equations (B.5)–(B.6) therefore give a uniform $L^{5/4}$ bound for the stopped family, implying uniform integrability. Removing the stopping identifies the Riccati expression with the true transform expectation. The finite discrete telescoping sum involves only finitely many integrable conditional expectations, to which the same moment-growth estimate applies.

Appendix C. Complete error control for inversion of the thirteen linear transforms

Lee [13] analyzes truncation and sampling errors in Fourier pricing. The bounds below use Gaussian decay and periodization moments for the thirteen profiles in this paper.

C.1 Damped inversion and the finite sum

For a positive measure μ, write $\widehat{\mu}(z) = \int e^{zy}\mu(dy)$. Take

$$\begin{aligned} \delta = 1/2, \qquad \Delta u = 1/5, \qquad N = 640, \\ U = N\Delta u = 128, \qquad L = 2\pi/\Delta u = 10\pi. \end{aligned}$$

When the negative moment is finite, $e^{-\delta y}F_\mu(y)$ is integrable and has Fourier transform $\widehat{\mu}(-\delta - iu)/(\delta + iu)$. Consequently,

$$F_\mu(y) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{e^{(\delta+iu)y}}{\delta + iu}\widehat{\mu}(-\delta - iu)\, du. \tag{C.1}$$

Substitute (3.6), and set $z_n = \delta + in\Delta u$, $w_0 = 1/2$, and $w_n = 1$ for $n \geq 1$. The coefficients in the finite real sum are

$$\begin{aligned} a_{0,n} &= \frac{e^{-r}\Delta u\, w_n}{\pi z_n}(95e^{z_n y_{95}} - 110e^{z_n y_{110}}), \\ a_{i,n} &= \frac{e^{-r}\Delta u\, w_n}{\pi z_n}\frac{100}{12}(e^{z_n y_{110}} - e^{z_n y_{95}}), \qquad 1 \leq i \leq 12. \end{aligned} \tag{C.2}$$

The corresponding transforms are $Ee^{-z_n Y}$ and $Ee^{Z_i - z_n Y}$. The zero frequency receives half weight, while the highest retained frequency receives full weight. The real part of the

remaining loading in a stock profile lies in $[-1/2, 1]$; its squared modulus requires real loadings in $[-1, 2]$.

C.2 Periodization error

Poisson periodization gives

$$\sum_{k\in\mathbb{Z}} e^{-\delta k L} F_\mu(y + kL).$$

For $k > 0$, use the total mass $M_{\mu,0}$. For $k = -j < 0$, use $F_\mu(y - jL) \le e^{y-jL} M_{\mu,-1}$. The sum of the nonzero-index terms is then at most

$$\frac{M_{\mu,0}}{e^{\delta L} - 1} + \frac{M_{\mu,-1} e^{y}}{e^{(1-\delta)L} - 1}. \tag{C.3}$$

Here $M_{0,0} = 1$, $M_{A,0} \le 100 e^{.01}$, $M_{0,-1} \le e^{.78}$, and $M_{A,-1} \le 100 e^{.78}$. Taking absolute values term by term in the price combination gives the periodization error allowance for a single law:

$$E_{\text{alias}} = e^{-.01} \frac{205 + 200 e^{.01} + e^{.78}(195 e^{y_{95}} + 210 e^{y_{110}})}{e^{5\pi} - 1}. \tag{C.4}$$

The difference between the two models is charged $2E_{\text{alias}}$. The frequency tails below ensure absolute summability of the Fourier coefficients, while (C.3) ensures absolute and uniform convergence of the periodization on compact intervals. Fourier uniqueness therefore identifies both representations with the same continuous function.

C.3 Frequency decay from the common Gaussian factor

Under the conditioning in (5.3), the total imaginary loading of each profile on the first-month factor U is $-u$. Set $K = E[e^{\text{real loading}} \mid \mathcal{G}]$. Conditional Gaussian integration, followed by Cauchy–Schwarz and conditional Jensen, gives

$$|\Phi_\nu(u)| \le E[K e^{-u^2 s I/2}] \le \sqrt{E e^{2\,\text{real loading}}} \sqrt{E e^{-s u^2 I}} \le \sqrt{M_2 L_\nu(u)}. \tag{C.5}$$

Here $L_\nu(u) = E_\nu e^{-s u^2 I}$. For the average of the twelve terms defining F_A, retain the additional price-level factor 100. Define

$$C_* = e^{-.01}\left(195 e^{\delta y_{95}} + 210 e^{\delta y_{110}}\right).$$

Under Q, $I \ge i_0 = h v_0$, so

$$L_Q(u) \le L_Q(U) e^{-s i_0 (u^2 - U^2)} \quad (u \ge U).$$

A validated upper bound for the true $L_Q(U)$ requires only 64 backward steps. Starting from $b = A = 0$, use

$$A \leftarrow A + dhb, \qquad b \leftarrow b + h(\alpha b^2 - \kappa b - s U^2). \tag{C.6}$$

At each level, verify $b \le 0$. The inequality $e^{bY^+} \le e^{bY}$ then ensures that this Gaussian recursion is an upper bound for the actual positive-part kernel. At the end, $\overline{L}_Q(U) = e^{A + b v_0}$. Bounding the discrete sum of a positive decreasing envelope by its integral gives

$$E_{\text{tail},Q} \le \frac{C_* \sqrt{M_2 \overline{L}_Q(U)}}{\pi s i_0 U^2}. \tag{C.7}$$

The integral inequality used here is

$$\int_U^\infty e^{-a(u^2-U^2)}\,\frac{du}{u} \le \frac{1}{2aU^2}.$$

For P, define

$$\Delta_U = \sqrt{\kappa^2 + 4\alpha s U^2}, \quad C_0 = \kappa/(2\alpha), \quad C_1 = C_0 + \Delta_U e^{-\Delta_U t_1}/\alpha,$$

$$C_P = \exp\{d(t_1 C_0 + \log 2/\alpha) + v_0 C_1\}, \qquad \gamma = \frac{1}{2}\sqrt{s/\alpha}(dt_1 + v_0).$$

The same CIR cross-ratio bound as in (A.7), now with terminal value $b = 0$, gives $L_P(u) \le C_P e^{-2\gamma u}$ for every $u \ge U$. Hence

$$E_{\text{tail},P} \le \frac{C_* \sqrt{M_2 C_P}}{\pi \gamma U} e^{-\gamma U}. \tag{C.8}$$

C.4 Branches of the continuous complex flow

Write $\kappa_q = \kappa - \rho\xi q$, $\gamma_q = (q^2 - q)/2$, and $D_q = \sqrt{\kappa_q^2 - 4\alpha\gamma_q}$. Divide each month into eight subintervals of length $\Delta t = 1/96$. The denominator of the Möbius flow is

$$\mathcal{D}_t = \cosh(D_q t/2) + (\kappa_q - 2\alpha b)\frac{\sinh(D_q t/2)}{D_q}.$$

The integral of the flow is

$$\int_0^{\Delta t} b(t)\,dt = \frac{\kappa_q \Delta t/2 - \log \mathcal{D}_{\Delta t}}{\alpha}. \tag{C.9}$$

On every subinterval, verify

$$\cosh(|D_q|\Delta t/2) - 1 + \frac{|\kappa_q - 2\alpha b|\Delta t}{2}\cosh(|D_q|\Delta t/2) < 1. \tag{C.10}$$

Absolute-value bounds on the power series imply $|\mathcal{D}_t - 1| < 1$ along the entire subinterval. The principal logarithm therefore remains continuously connected to the initial value 1 of the denominator. At the boundary $\operatorname{Re} b = 1$, the imaginary-part quadratic form

$$-\alpha(\operatorname{Im} b)^2 - \rho\xi(\operatorname{Im} q)(\operatorname{Im} b) - (\operatorname{Im} q)^2/2$$

is nonpositive and the remaining real drift is strictly negative. Thus the continuous flow satisfies $\operatorname{Re} b \le 1$. Appendix B.3 identifies the affine expression with the true transform.

C.5 Combination and precision

Let $\hat{e}_L$ be the finite proxy difference. The true linear difference is enclosed by

$$\hat{e}_L + [-E_L, E_L], \qquad E_L = 2E_{\text{alias}} + E_{\text{tail},P} + E_{\text{tail},Q} + \delta_Q \sum_{n,i} |a_{i,n}|. \tag{C.11}$$

All finite sums and coefficients are evaluated as Arb enclosures. Substituting the exact rational endpoints gives

$$E_L \le .00059423805983999257450 5.$$

Combining this with the asymmetric payoff remainder obtained from (A.9), according to Theorem 4.1, yields (4.4).

Appendix D. Gaussian density expansions and finite-step bounds

D.1 Variance expansion

For $t/h \in \mathbb{N}$, set

$$x = -\frac{t}{h}\log(1-\kappa h) - \kappa t \geq 0.$$

The power series for the logarithm gives

$$0 \leq x - \frac{\kappa^2 t h}{2} \leq \frac{\kappa^3 t h^2}{3(1-\kappa h)}, \qquad x \leq \frac{\kappa^2 t h}{2(1-\kappa h)}.$$

Using $|1 - e^{-x} - x| \leq x^2/2$ and substituting

$$I_h(t) - I(t) = \frac{D}{\kappa} e^{-\kappa t}(1 - e^{-x})$$

yields $|I_h(t) - I(t) - h a(t)| \leq h^2 B_I(t)$. Adding the two cumulative remainder bounds at adjacent times gives (6.5).

D.2 Density derivatives

For a single increment, let $X = Y - r/12$. Its density is

$$f_s(y) = (2\pi s)^{-1/2} \exp[-X^2/(2s) - X/2 - s/8].$$

Hence

$$\partial_s \log f_s = -\frac{1}{2s} + \frac{X^2}{2s^2} - \frac{1}{8}, \qquad \partial_s^2 \log f_s = \frac{1}{2s^2} - \frac{X^2}{s^3}.$$

With $X = -s/2 + \sqrt{s}G$, the first-order score has mean zero and second moment $J(s) = 1/(2s^2) + 1/(4s)$. Moreover,

$$E|\partial_s^2 \log f_s| \leq \frac{3}{2s^2} + \frac{1}{4s}.$$

Independence of the coordinates makes the cross-score expectations vanish, yielding the two L^1 derivative bounds used in the proof of Theorem 6.1. On a variance box with a positive lower bound and a finite upper bound, the first and second density derivatives are uniformly dominated by $C(1+|y|^4)e^{-c|y|^2}$. Difference quotients, parameter integrals, and payoff integrals may therefore be interchanged as required.

D.3 A complete finite-step comparison

For two variance vectors $s, \tilde{s}$, suppose that their endpoints and the segment joining them satisfy $s_j, \tilde{s}_j \geq u_j > 0$. Integrating the first-order score along the segment gives

$$\mathrm{TV}(p_s, p_{\tilde{s}}) \leq \left\{ \sum_j (\tilde{s}_j - s_j)^2 \left(\frac{1}{8u_j^2} + \frac{1}{16 u_j} \right) \right\}^{1/2}. \tag{D.1}$$

For a payoff with range length L, the price difference is at most $Le^{-r\tau}$ TV. On (6.10), using the uniform bounds $|D| \le .015$ and $u_j \ge .03/12$, finite computation gives

$$|e_{h,\mathrm{Asian}}| \le .004990923469, \qquad |R_{h,\mathrm{Asian}}| \le .000165216416.$$

The posterior module uses the tighter actual minimum variance for each month, so its reference bound ε_0 at fixed parameters can be smaller than this uniform bound.

D.4 An effectively computable representation of the leading coefficient

For rational parameters and the explicit Lipschitz payoffs considered here, the leading coefficient in (6.7) is an effectively computable integral. Let

$$S_d(G) = \sum_j d_j \left[\frac{G_j^2 - 1}{2s_j} - \frac{G_j}{2\sqrt{s_j}}\right], \qquad F = \sum_j d_j^2 J(s_j).$$

After centering the payoff, omitting the region $\max_j |G_j| > R$ incurs error at most

$$e^{-r\tau}\frac{L}{2}\sqrt{24F}\, e^{-R^2/4}. \tag{D.2}$$

This follows directly from Cauchy–Schwarz and a union bound for Gaussian tails. On $[-R,R]^{12}$, define

$$M_S = \sum_j |d_j| \left[\frac{R^2+1}{2s_j} + \frac{R}{2\sqrt{s_j}}\right],$$
$$L_S = \sum_j |d_j| \left[\frac{R}{s_j} + \frac{1}{2\sqrt{s_j}}\right].$$

If the payoff has Lipschitz constant L_z as a function of the fixing log-price vector, the integrand has Lipschitz constant at most

$$L_H = L_z\left(\sum_j \sqrt{s_j}\right) M_S + \frac{L}{2} L_S.$$

The sum of box-midpoint values weighted by exact Gaussian box probabilities, for boxes of side length ℓ, has total interior error at most $e^{-r\tau} L_H \ell/2$. Choosing R, ℓ together with (D.2) therefore gives a finite algorithm with at most $\lceil 2R/\ell \rceil^{12}$ boxes. This representation establishes computability; Section 9 distinguishes this representation from an actually executed high-precision evaluation of the leading coefficient.

Appendix E. Perturbation constants for positive volatility of volatility

In the continuous model, $D_t = V_t^\xi - v(t)$ admits a linear variation-of-constants representation. Since $EV_t^\xi = v(t) \le M$, the Itô isometry gives

$$ED_t^2 = \xi^2 \int_0^t e^{-2\kappa(t-s)} EV_s^\xi \, ds \le \frac{\xi^2 M}{2\kappa_-}. \tag{E.1}$$

The discrete deterministic reference remains nonnegative. The positive-part map is 1-Lipschitz, and the Gaussian innovation has conditional mean zero. Thus, writing $D_j^{(2)} = E(V_j^\xi - v_j)^2$, we obtain

$$D_{j+1}^{(2)} \le (1-\kappa h)^2 D_j^{(2)} + \xi^2 h E V_j^\xi \le (1-\kappa h)^2 D_j^{(2)} + \xi^2 h (M + \sqrt{D_j^{(2)}}). \tag{E.2}$$

Take $C_Q^2 = 2M/\kappa_-$. Condition (7.1) gives $\xi C_Q \le M$ and $1-(1-\kappa h)^2 \ge \kappa_- h$. Induction from the zero initial error then yields $D_j^{(2)} \le \xi^2 C_Q^2$. This step retains the effect of projection on EV_j^ξ.

For the deterministic reference $v \ge m$,

$$|\sqrt{x} - \sqrt{v}|^2 = \frac{|x-v|^2}{(\sqrt{x}+\sqrt{v})^2} \le |x-v|^2/m.$$

The drift contribution to the log-stock difference is bounded by $T\xi C_\ell/2$. The Itô isometry and Doob's L^2 inequality bound the martingale contribution by $2\xi C_\ell \sqrt{T/m}$, proving (7.3). Correlation between the stock Brownian motion and the variance driver does not alter these inequalities for adapted stochastic integrals.

For a put, the absolute derivative with respect to log stock price is at most K on the active region. For the capped Asian, consider a line segment in the fixing log-price vector. On the active region $K_1 < A < K_2$, the gradient has ℓ^1-norm $A \le K_2$; elsewhere it is zero. Piecewise absolute continuity gives the global Lipschitz constant 110. Thus, for maturity T,

$$e_{\ell,i} = K_i e^{-rT_i} \xi_{\max} C_\ell \left(T_i/2 + 2\sqrt{T_i/m} \right),$$

$$e_{\ell,J} = 110 e^{-r} \xi_{\max} C_\ell \left(1/2 + 2/\sqrt{m}\right). \tag{E.3}$$

In (7.5), $m = .03, M = .06, \kappa = 3$, so $C_P = .1, C_Q = .2$, giving

$$e_{P,J} \le .000131198492451391, \qquad e_{Q,J} \le .000262396984902781. \tag{E.4}$$

For (7.4), first enclose the reference variance difference over 256 closed intervals in κ using outward arithmetic. Then apply

$$|p_h^\xi - p_c^\xi| \le |p_h^0 - p_c^0| + 110\xi(C_P + C_Q)(1/2 + 2/\sqrt{m})$$

to obtain the full-box price bound stated in the main text. The remaining parameters are enclosed through uniform bounds on $m, M, |v_0 - \bar{v}|$, rather than by substituting endpoint samples for full enclosures.

Appendix F. Posterior inputs, normalization, and finite coupling

F.1 Exact quotes

The quotes are ordered by $T = 1/4, 1/2, 1$, with $K = 90, 100, 110$ within each maturity. The following rational numbers are taken as the given synthetic inputs:

$$y = \begin{pmatrix} 526706580970763/562949953421312 \\ 285901443526503/70368744177664 \\ 754049251176359/70368744177664 \\ 580752698287259/281474976710656 \\ 794307614955649/140737488355328 \\ 830842010134081/70368744177664 \\ 269207942741719/70368744177664 \\ 137257851354697/17592186044416 \\ 1918003280297167/140737488355328 \end{pmatrix}. \tag{F.1}$$

These fractions are the exact binary64 values in the reference input file; the single-quote case uses the eighth entry. They are inputs to the likelihood, and the auxiliary computation that produced them need not be regarded as a proof of continuous-model prices.

F.2 Reference prices and target constants

Let

$$b_P(t) = \frac{1 - e^{-3t}}{3}, \quad b_Q(t) = \frac{1 - (1 - 3h)^{t/h}}{3},$$
$$I_m(T; u) = .045T + (u - .045) b_m(T).$$

The reference put price is computed by the exact Black formula

$$\begin{aligned} p_m(K, T; u) &= K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1), \\ d_1 &= \frac{\log(S_0/K) + rT + I_m/2}{\sqrt{I_m}}, \qquad d_2 = d_1 - \sqrt{I_m}. \end{aligned} \tag{F.2}$$

Since $b_m(T) > 0$ and vega is positive, the reference price over each entire u-cell is enclosed by its endpoint values. Expanding both endpoints by (E.3) then encloses the true prices over the entire cell and all ζ.

Let

$$\begin{aligned} a_{m,j} &= b_m(t_j) - b_m(t_{j-1}), \\ z_{m,j}(u) &= .045/12 + (u - .045)a_{m,j}, \qquad B_* = 15e^{-.01}. \end{aligned}$$

The centered score bound in Appendix D gives

$$L_0 = \frac{B_*}{2}\sqrt{\sum_j a_{P,j}^2 J(z_{P,j}(.03))},$$

$$\begin{aligned} \varepsilon_0 &= \frac{B_*}{2}\sqrt{\sum_j \Delta_j^2 J(\underline{z}_j)}, \\ \Delta_j &= .015|a_{Q,j} - a_{P,j}|, \qquad \underline{z}_j = \min(z_{P,j}(.03), z_{Q,j}(.03)). \end{aligned} \tag{F.3}$$

F.3 Likelihood enclosures over entire cells

For $d = 1$ or 9,

$$\Sigma_d^{-1} = \frac{64}{27}\left(I_d - \frac{\mathbf{1}\mathbf{1}^T}{d+3}\right).$$

Writing the residual vector as r, its potential is a sum of nonnegative terms:

$$\frac{1}{2}r^T\Sigma_d^{-1}r = \frac{32}{27(d+3)}\left(3\sum_i r_i^2 + \sum_{i<j}(r_i - r_j)^2\right). \tag{F.4}$$

Evaluate this expression and its exponential outward, then round to the exact integer lattice with spacing 2^{-96}, to obtain

$$0 < l_{m,i} \le w_m(u, \zeta) \le u_{m,i} \le 1$$

throughout the entire cell and nuisance-parameter domain. Integration against the probability measure $\nu(d\zeta)$ gives the same enclosure for $\bar{w}_m(u)$.

F.4 Shared normalization and inverse-CDF pairing

Each cell has the same prior mass. At a node x_k, denote the sums of lower and upper mass bounds to its left by $L_<, U_<$, and those to its right by $L_>, U_>$. The marginal CDF G_m satisfies

$$\underline{G}_m(x_k) = \frac{L_<}{L_< + U_>}, \qquad \overline{G}_m(x_k) = \frac{U_<}{U_< + L_>}. \tag{F.5}$$

The function $A/(A+B)$ increases in A and decreases in B, so these enclosures retain the shared mass in the numerator and denominator. The normalizing constant is also enclosed directly by summing all weights.

Let U be a common uniform random variable and set $U_m = G_m^{-1}(U)$. If $U_m \in [x_i, x_{i+1}]$, then

$$U \in [\underline{G}_m(x_i), \overline{G}_m(x_{i+1})]. \tag{F.6}$$

For each P-cell, enumerate the leftmost and rightmost indices j_-, j_+ of all Q-cell intervals whose corresponding ranges in (F.6) can intersect its range. Set

$$\delta_u = \Delta x \max_i \max\{|i - (j_+ + 1)|,\ |i + 1 - j_-|\}, \quad \Delta x = .03/4096. \tag{F.7}$$

The true cell pair must lie in this candidate set; hence $|U_P - U_Q| \le \delta_u$ almost surely. All comparisons use exact integers and fractions.

The enclosures of the normalizing constants for the true three-dimensional prior, obtained from all cells, are shown in Table F.1.

Table F.1. Posterior normalizing-constant enclosures on 4096 cells.

Case	Z_P	Z_Q
Single quote	[.935743002637584884, .935913179433726711]	[.935667223967444023, .935927287640792653]
Nine quotes	[.620128921272339539, .621155228251382142]	[.619743550733119263, .621256330676942526]

The target radii from Appendix E, together with (F.3) and (F.7), yield (7.8). This procedure encloses each model's own full posterior.

Appendix G. Reproducible implementation

Start with the `V_20261101` tag at https://github.com/130U/certified-valuation-arithmetic-asian-options/tree/V˙20261101. It contains the manuscript, step-grid and posterior-grid experiments, the second Heston point, and the deterministic coupling example. The root `PUBLICATION-MANIFEST.json` lists the files in this checkout and their hashes; `code/MANIFEST.json` identifies the current numerical package.

The execution archive described below contains the source copies, inputs, results and receipts supporting the calculations in this publication.

Table G.1. Numerical modules and their results.

Module	Corresponding result
`asian`	Theorem 4.1 and Corollary 4.2 at the principal parameter point.
`tt`	Theorem 6.1 and constants on the deterministic face (6.10).
`small-xi`	The five-dimensional perturbation box (7.4).
`posterior-zero`	Gaussian-reference posterior and target constants.
`posterior`	The restricted prior (7.5) and Theorem 7.1.

The reference runtime is unoptimized CPython 3.12 on Windows x86-64 with `python-flint==0.8.0`. From the publication checkout, the commands below create the environment, verify the current package and execute the principal Asian certificate with both implementations. Each double hyphen is a literal command-line character.

```
py -3.12 -m venv .venv
./.venv/Scripts/python.exe -m pip install -r code/requirements.txt
./.venv/Scripts/python.exe code/run.py verify
./.venv/Scripts/python.exe code/run.py run --module asian --independent
```

The run prints its new directory under `code/runs/`. Replace `run-ID` below with that directory name. To execute every listed module, replace `asian` in the preceding command with `all`.

```
./.venv/Scripts/python.exe code/run.py check --run code/runs/run-ID
```

Installation may require network access. The numerical calculations do not. No dependency wheel is included; offline installation requires an authentic compatible wheel obtained separately. Assertions enforce mathematical conditions and resource contracts. Each worker uses one thread and a 256 MiB hard memory limit.

The completed calculations are preserved in `code/evidence/executions.zip`. The archive and its member inventory are identified by `code/evidence/receipt.json`. To verify the saved evidence, use a fresh directory name and run

```
python code/verification/read_evidence.py --directory evidence-check --output evidence-readback.json
```

This standard-library check reads the archive, validates its inventory and reconciles the exact result ledgers. It does not rerun the pricing calculations. `docs/EVIDENCE.md` lists the calculation programs, results and fresh execution commands. Saved-result verification and fresh numerical execution are distinct operations.

The step-grid ledger `code/results/step-grid.json` maps each printed numerical row to the exact source field, result, hash and receipt. The principal weighted moments are in `code/reference/asian-remainder-result.json`; the linear center, transform contributions and signed price enclosure are in `code/reference/asian-linear-result.json`. Exact synthetic observations are in `code/data/synthetic_quotes.json`. Appendices A, B and C describe the nonlinear catalog, original projected recurrence and linear inversion, respectively.

The reference comparison includes every numerical JSON field, including arrays and rational endpoints. Only hash metadata and six named elapsed-time fields are excluded. The alternative constructions described below share Arb and the analytical inequalities; numerical agreement verifies the implemented finite calculations within those assumptions, without supplying an independent derivation of the common inequalities.

At 192 Euler steps, the projection exponent does not exceed 8, whereas the projection argument requires it to be strictly greater than 8. The argument therefore supplies no certificate for that grid. The execution records distinguish this analytical exclusion from numerical acceptance targets and resource stops. Reported task times are machine- and load-specific.

G.1 Numerical constructions and recorded cost

The nonlinear-moment module evaluates 24,388 continuous real-flow blocks and 815,360 discrete real-recursion steps at the principal point. The linear module evaluates 641 frequencies and thirteen profiles, for 6,399,744 discrete steps and 799,968 continuous branch checks. The principal programs use 256-bit Arb [12]; the check programs use 384-bit Arb with different numerical constructions.

For the weighted remainder, `check-asian-remainder.py` replaces the principal root/cross-ratio Riccati formula by a hyperbolic two-by-two flow, constructs the loading catalog from ordered pairs, and separately indexes the reverse discrete grid. It accumulates the projection prefix as a product rather than exponentiating a sum. For linear inversion, `check-asian-linear.py` replaces the principal real/imaginary discrete recursion by a complex `acb` recursion and expresses the continuous hyperbolic blocks through exponentials. It verifies every frequency contribution and recomputes the finite error allowances without calling the principal numerical functions. The checks share Arb, the analytic identities, and the theoretical bounds. They test implementation consistency and arithmetic stability rather than supplying independent derivations of those bounds. The root-flow and matrix-flow calculations also enclose the same grouped positive moments used in (4.11).

Recorded worker times for the nonlinear and linear modules were approximately 12.02 and 33.20 seconds, respectively, with one thread and a 256 MiB hard memory limit per process. In the step-grid study, the summed outer wall times for the five Asian tasks were approximately

122.3, 184.0, and 431.6 seconds at 384, 768, and 1536 steps. These latter observations include controllers and check programs under the recorded execution load; they are not portable performance guarantees. Each receipt retains its job times, operation counts, and resource contract.

G.2 Sampling illustrations and posterior marginal envelopes

The Asian coefficient illustration uses binary64 arithmetic and 2,097,152 Gaussian increment vectors per point. It evaluates the exact deterministic-variance Gaussian coefficient integrand and density-ratio residual with common vectors across the four step sizes. Subtracting the known zero-mean score and using `expm1` reduces cancellation in the scaled residual. Inputs, seeds, formulas, sampling estimates, and estimated standard errors are recorded in `code/results/asian-beta-diagnostic.json`. Its normal-approximation sampling intervals are distinct from the analytic coefficient enclosures and deterministic remainder bounds in Section 6.

The separate NumPy path study uses 32,768 paths for each of fourteen cases, coupling the original projected Euler grid to a grid with twice as many steps. The file `code/results/path-diagnostics.json` records seeds, binary64 inputs, sampled prices, standard errors, and negative-proposal counts. Its fine law retains discretization bias, so these sampling results do not enclose the continuous-model price.

The posterior grid summary, `code/results/posterior-grid.json`, retains full rational CDF envelopes and the 2.5, 50, and 97.5 percent quantile brackets of the initial-variance marginal. The absolute Asian-price target quantiles are not part of that computation.

G.3 Second-point and coupled-moment reproduction

The second Heston point in Section 8.3 and the deterministic coupling example in Appendix H are supplied with their programs, inputs, results and execution evidence in the publication package. To recompute the second point in the same Windows environment, choose a new output directory and run

```
./.venv/Scripts/python.exe code/experiments/asian_parameter_point.py prepare --directory new-point
./.venv/Scripts/python.exe code/experiments/asian_parameter_point.py run --directory new-point
```

The run writes its result capsule in the new directory. The point configuration changes only v_0 to $1/25$. The preparation checks the current numerical manifest and constructs explicit parameter copies. The weighted, linear, pilot and two alternative-construction jobs recompute all affected quantities. Declared worker budgets are 120 seconds for each weighted calculation, 360 for each linear calculation and 30 for the pilot; the one-thread and 256 MiB limits remain fixed. The recorded five-job outer time is approximately 97.5 seconds. The source copies, results and execution records are included in `code/evidence/executions.zip`; its receipt identifies the complete member inventory. The exact six-component width ledger is in `code/results/point-v004-result.json`.

The deterministic coupling example uses finite closed Gaussian moments at 384 and 512 bits. Its fresh calculation command is

```
./.venv/Scripts/python.exe code/experiments/coupling_example.py --output coupling.json
```

The exact result is `code/results/coupling-v004-result.json`. Appendix H gives the proof, and `docs/COMPUTATION.md` describes the finite moment evaluation. The publication and numerical manifests record the source and result identities. The calculation verifies the nonlinear component on $\xi = 0$; it does not replace the positive-ξ pricing calculation.

Appendix H. A direct coupled-remainder inequality

The following sufficient inequality preserves cancellation between the two conditional remainders under a coupling of their conditional variables.

Proposition H.1 (A remainder bound under a coupling). Fix $K > 0$ and $c > 0$, assume that both marginal laws satisfy $\mathsf{H}_s(c)$, so that the remainders are finite, and couple conditional triples (a_P, b_P, σ_P), (a_Q, b_Q, σ_Q), where $b_\nu = cg_\nu$, all entries are positive, and each marginal is the conditional triple in (2.6). Set

$$s_- = \min(\sigma_P, \sigma_Q), \qquad s_+ = \max(\sigma_P, \sigma_Q), \qquad z = \max(|a_P - b_P|, |a_Q - b_Q|),$$

$$M_K = \frac{e^{2s_+^2}}{Ks_-\sqrt{2\pi}}, \qquad N_K = \frac{e^{2s_+^2}(1 + 4/e + 8s_+^2)}{Ks_-^2\sqrt{2\pi}},$$

$$\Xi_K = M_K z\,(|a_Q - a_P| + |b_Q - b_P|) + \frac{1}{2} N_K z^2 |\sigma_Q - \sigma_P|.$$

If $E\Xi_K \le B_K < \infty$, then

$$|R_{K,Q} - R_{K,P}| \le B_K. \tag{H.1}$$

In particular, writing $B = e^{-rT}(B_{K_1} + B_{K_2})$, the exact nonlinear contribution to the call-spread price difference belongs to $[-B, B]$. This interval may be intersected with the nonlinear interval in (4.2) before adding the linear-transform interval. Under the diagonal coupling for identical conditional laws, $\Xi_K = 0$.

Proof. Let Z be standard normal and define

$$F_K(x; s) = E[(xe^{sZ} - K)^+], \qquad r_K(a, b, s) = F_K(a; s) - F_K(b; s) - (a - b)\partial_x F_K(b; s).$$

Conditional integration gives $R_{K,\nu} = E r_K(a_\nu, b_\nu, \sigma_\nu)$. For $x, s > 0$, with $y = \log(x/K)$,

$$H_K(x, s) := \partial_x^2 F_K(x; s) = \frac{K}{sx^2\sqrt{2\pi}} e^{-y^2/(2s^2)}, \qquad \partial_s H_K = \frac{H_K}{s}\left(\frac{y^2}{s^2} - 1\right).$$

Completing the square gives

$$H_K(x, s) = \frac{e^{2s^2}}{Ks\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{y}{s} + 2s\right)^2\right].$$

For $w = y/s + 2s$,

$$\left|(w - 2s)^2 - 1\right| e^{-w^2/2} \le (2w^2 + 8s^2 + 1)e^{-w^2/2} \le 4/e + 8s^2 + 1.$$

Therefore, whenever $s \in [s_-, s_+]$, $0 \le H_K \le M_K$ and $|\partial_s H_K| \le N_K$, uniformly in $x > 0$. Differentiation of the remainder and Taylor's integral formula give

$$\partial_a r_K = \partial_x F_K(a; s) - \partial_x F_K(b; s), \qquad \partial_b r_K = -(a - b)H_K(b, s),$$

$$\partial_s r_K = (a - b)^2 \int_0^1 (1 - t)\partial_s H_K(b + t(a - b), s)\, dt.$$

Interpolate the three entries linearly between the coupled endpoints. All interpolated a, b, s stay positive, $|a - b| \le z$, and $s \in [s_-, s_+]$. The fundamental theorem of calculus and the displayed derivative bounds yield, for each coupled pair,

$$|r_K(a_Q, b_Q, \sigma_Q) - r_K(a_P, b_P, \sigma_P)| \le \Xi_K.$$

Taking expectations proves (H.1). Applying the triangle inequality to the two strikes and multiplying by the discount proves the price-unit statement. □

The s_-^{-2} factor requires coupled-moment control beyond (2.7) and the marginal affine catalog. If a family of verified couplings satisfies $E\Xi_K \to 0$, then (H.1) gives vanishing remainder differences. No such convergence bound or rate for the positive-volatility-of-volatility Heston example is asserted here.

H.1 An executed coupling example on the deterministic-variance face

At $\xi = 0, \kappa = 3, \bar{v} = 9/200, v_0 = 1/25, S_0 = 100, r = 1/100, T = 1, h = 1/768$, retain the twelve monthly fixings, the call spread with strikes 95 and 110, and $c = 1254433/1250000$. Write $s_{\nu,j} = I_\nu(t_j) - I_\nu(t_{j-1})$, using the exact continuous and Euler cumulative variances in (6.2). Couple the laws by twelve common independent standard normals. With $\sigma_\nu = \sqrt{s_{\nu,1}}$, remove the first centered normal factor and put

$$m_{\nu,i} = rt_i - I_\nu(t_i)/2, \qquad \ell_{\nu,i,j} = \sqrt{s_{\nu,j}}\mathbf{1}_{\{j\le i\}}, \quad 2 \le j \le 12,$$

$$a_{\nu,i} = 100e^{m_{\nu,i}+\ell_{\nu,i}\cdot Z}, \qquad a_\nu = \frac{1}{12}\sum_i a_{\nu,i}, \qquad g_\nu = 100e^{\bar{m}_\nu+\bar{\ell}_\nu\cdot Z}, \qquad b_\nu = cg_\nu.$$

Here $Z = (Z_2, \dots, Z_{12})$, and bars denote averaging over fixing index $i = 1, \dots, 12$. The drift includes the first-month term $-s_{\nu,1}/2$; hence the removed factor has the centered Gaussian law required by $\mathsf{H}_s(c)$. Here the stock driver $\rho W + \sqrt{1-\rho^2}B$ is combined into a single Brownian motion. On $\xi = 0$ the variance is deterministic and the price law is independent of ρ; hence $\sigma_\nu^2 = s_{\nu,1}$, using the full first-month variance. The separate-law comparison below uses this same conditioning.

All conditional second moments, including mixed P/Q moments, follow from

$$E\big[100e^{m+u\cdot Z}\,100e^{n+v\cdot Z}\big] = 10000e^{m+n+\|u+v\|^2/2}.$$

Define

$$q_\nu = E(a_\nu - b_\nu)^2, \quad Z_2 = q_P + q_Q, \quad D_a = E(a_Q - a_P)^2, \quad D_b = E(b_Q - b_P)^2.$$

For the deterministic M_K, N_K in the coupling proposition, the pointwise bound $z^2 \le (a_P - b_P)^2 + (a_Q - b_Q)^2$ and Cauchy–Schwarz give the executable upper bound

$$E\Xi_K \le B_K := M_K\sqrt{Z_2}(\sqrt{D_a} + \sqrt{D_b}) + \frac{1}{2}N_K Z_2|\sigma_Q - \sigma_P|.$$

The finite Gaussian-moment computation at 384 and 512 bits yields

$$E\Xi_{95} \le .000023887739528734380066, \qquad E\Xi_{110} \le .000020630320502088782784.$$

Consequently the discounted nonlinear spread contribution belongs to the symmetric interval with radius $e^{-r}(B_{95} + B_{110})$, whose width is at most .000088150195864703911227. For the same laws, conditioning, and scale, the separate-law estimate has width in

$$[.015383637151510699801154,\ .015383637151510699801155].$$

The ratio of widths is below .005731. Strict improvement is checked using the outward upper endpoint of the coupled width and the lower endpoint of the separate width. The comparison uses $D_\nu = e^{2\sigma_\nu^2}q_\nu/\sigma_\nu$, so all weighted moments refer to the same Gaussian factor.

This is a closed-moment enclosure of the coupled remainder expectations, with no Monte Carlo or unbounded-domain quadrature. It establishes narrowing of the nonlinear component at one deterministic-variance point. It does not compute a complete price interval, a signed

Asian leading coefficient, or a coupled certificate for $\xi > 0$. The two precisions and two algebraic Gaussian-moment expressions share the analytical proof and Arb; they are arithmetic consistency checks.

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